

Transported Complier Effects with Weak First Stages

CausalSmith

August 16, 2026

Abstract

This paper studies honest confidence sets for a transported complier average causal effect in a two-sample encouragement design. A source population supplies an instrumental-variables experiment with noncompliance, while a target population supplies the covariate law. Under the transported-IV model class, with source contrasts satisfying the stated measurability and integrability conditions, the transported Wald ratio equals the target complier contrast and lies in the compact range $[-1, 1]$. The paper characterizes the oracle minimax expected-length frontier for honest confidence sets over transported triangular arrays satisfying the stated overlap, sample-size, envelope, second-moment, and degrading-array conditions, with weight second moment κ_n and transported first-stage mean μ_n . The frontier is indexed by the effective strength

$$t_n = \frac{n\mu_n^2}{\kappa_n}.$$

For every fixed threshold $t_0 > 0$, oracle procedures that know the transport weight and source propensity have minimax expected length of order

$$\min\{1, t_0^{-1/2}\}.$$

A transported Anderson–Rubin score-inversion set attains this order. The unit-transport-weight least-favorable slice is embedded in the transported class, so its weak-ratio lower bound applies to the larger transported model. Up to constants depending on the sampling ratio, overlap, and noncoverage level, the same order holds conditionally on any admissible fixed covariate-shift geometry. In finite-cell designs, target cell frequencies and a collision-scale U-statistic yield feasible score-inversion procedures that learn the target weights from the target covariate sample and attain the same order under the stated cell-growth and overlap conditions.

1 Introduction

Encouragement experiments with imperfect take-up identify complier effects in the source population under the usual instrumental-variables restrictions [Bloom, 1984, Imbens and Angrist, 1994, Angrist et al., 1996]. Many applications also require a target-population interpretation: the source experiment is observed in one population, while the policy-relevant covariate distribution belongs to another. Transportability and generalizability methods address this covariate-shift problem through support and weighting conditions [Joseph Hotz et al., 2005, Cole and Stuart, 2010, Stuart et al., 2011, Tipton, 2013, Dahabreh et al., 2020b,a, Nguyen et al., 2017, Pearl and Bareinboim, 2011, Bareinboim and Pearl, 2013, 2016]. For transported noncompliance, the estimand is a ratio: a transported reduced form divided by a transported first stage. This ratio

structure makes weak identification central when the target complier share is small or when transport weights are dispersed.

This paper characterizes honest expected length for that transported complier ratio. The source sample contains (X, Z, D, Y) from an encouragement design, and the target sample contains covariates X . The target-to-source density ratio $w = dP_T/dP_S^X$ transports source conditional contrasts to the target covariate law. Under the transported-IV model class $\mathcal{P}_n$ in [definition 3](#), with source contrasts satisfying the stated measurability and integrability conditions, [theorem 1](#) shows that the transported reduced form equals the target complier-weighted outcome contrast, the transported first stage equals the target complier share, and their ratio equals the target complier effect θ_T . Bounded outcomes and binary treatment receipt place θ_T in $\Theta = [-1, 1]$, so expected length is measured on a fixed compact parameter space.

The main statistical object is the effective strength

$$t_n = \frac{n\mu_n^2}{\kappa_n},$$

where μ_n is the transported first-stage mean and $\kappa_n = \mathbb{E}_S[w(X)^2]$ is the Kish-type dispersion scale of the transport weights. This scalar combines two sources of difficulty. A small μ_n is the familiar weak-first-stage problem from IV ratio inference [[Anderson and Rubin, 1949](#), [Fieller, 1954](#), [Staiger and Stock, 1997](#), [Stock et al., 2002](#), [Moreira, 2003](#), [Andrews et al., 2006](#), [Mikusheva, 2010](#), [Andrews et al., 2019](#)]. A large κ_n records the variance inflation from transporting the source experiment to a different target covariate law. The frontier conditions allow $k_n \rightarrow \infty$, $k_n = o(\sqrt{n})$, and $\mu_n \rightarrow 0$, so the analysis covers triangular arrays in which covariate shift grows while the transported first stage weakens.

The oracle benchmark allows procedures to use the transport weight w and the source propensity e . [Theorem 2](#) proves that every oracle honest confidence-set sequence over $\mathcal{P}_n$ has limiting worst-case expected length at least a constant times

$$\min\{1, t_0^{-1/2}\}$$

over laws satisfying $t_n \geq t_0$. The same lower bound already holds on the uniform finite-cell submodel $\mathcal{N}_n$. [Theorem 3](#) gives the matching construction: invert the transported residual score

$$\frac{1}{n} \sum_{i=1}^n w(X_i) \left\{ \frac{Z_i}{e(X_i)} - \frac{1 - Z_i}{1 - e(X_i)} \right\} (Y_i - \vartheta D_i)$$

with a radius proportional to $\sqrt{\hat{\kappa}_n/n}$. This Anderson–Rubin style inversion is honest over $\mathcal{P}_n$ and attains the same expected-length order for every fixed $t_0 > 0$.

The fixed-geometry results isolate the role of covariate shift. For an admissible deterministic geometry $\mathfrak{g} = (P_S^X, w, e)_{n \geq 1}$, [definition 16](#) holds the source covariate law, target weight, and propensity fixed while allowing the causal response law to vary. [Theorem 4](#) establishes the same two-sided order

$$V_{\mathfrak{g}}^*(t_0) \asymp \min\{1, t_0^{-1/2}\}$$

with constants uniform over admissible geometries. The lower bound uses the least-favorable family in [definition 19](#): the complier probability is proportional to the transport weight, so the transported first stage is calibrated to t_0 , and a local complier-outcome tilt moves the target ratio while the observed-data laws remain close in total variation as recorded in [definition 23](#).

Theorem 5 specializes this comparison to $w = 1$, where $\kappa_n = 1$, $P_T = P_S^X$, and the effective strength reduces to $n\mu_n^2$.

The feasible results replace oracle-known weights by target-sample learning in finite-cell designs. In the uniform finite-cell class $\mathcal{N}_n$, the source covariate law assigns mass $1/k_n$ to each cell and the source propensity equals $1/2$. **Theorem 6** constructs a sample-only score-inversion set using target empirical cell frequencies, source cellwise reduced-form and first-stage contrasts, and a target-sample U-statistic for the collision scale. Under $N_n/n \rightarrow c > 0$ and $k_n = o(\sqrt{n})$, this feasible procedure is honest over $\mathcal{N}_n$ and attains the oracle frontier order. **Theorem 7** extends the construction to regular finite-cell designs with known source cell probabilities bounded between fixed multiples of $1/k_n$ and known cell-varying propensities satisfying overlap.

The contribution is therefore a matched minimax expected-length characterization for transported complier ratios under the stated transported-IV, overlap, positivity, support, envelope, dispersion, sample-size, and cell-growth conditions. The analysis connects transported noncompliance methods such as [Rudolph and Laan \[2016\]](#) and [Chen and Huang \[2025\]](#) with weak-identification-robust LATE and two-sample IV inference [[Choi et al., 2018](#), [Choi and Shen, 2019](#), [Ma, 2023](#), [Smucler et al., 2025](#)]. For the compact transported complier ratio under the stated transported-IV restrictions and growth conditions, the frontier is indexed by $n\mu_n^2/\kappa_n$, so first-stage weakness and transport-weight dispersion enter through a common scale.

The appendix gives the proof details and a verification note for the checked mathematical statements; see [section 6](#).

2 Setup, Transport, and Honest Risk

At row n , the two-population full-data potential-outcome world is governed by P_n^F on S , X , $D(0), D(1)$, and $Y(0), Y(1)$. The source population, $S = 1$, supplies an encouragement experiment with observed variables Z , D , and Y ; the target population, $S = 0$, supplies the covariate law to which the complier effect is transported. The measurable covariate space is $\mathcal{X}$. Potential outcomes are bounded because the expected-length problem is posed over a compact causal range, and treatment receipt is binary because compliance type is the margin through which the encouragement identifies the complier contrast.

Definition 1. `[synth_1]` (Inverse-propensity instrument score $H(O)$) *For a source observation $O = (X, Z, D, Y)$ and source propensity e , the inverse-propensity instrument score is the function H given by*

$$H(O) = \frac{Z}{e(X)} - \frac{1 - Z}{1 - e(X)}.$$

The score in [definition 1](#) is the source inverse-propensity contrast used to recover conditional reduced-form and first-stage contrasts under the source experiment. Its normalization keeps the assignment contrast centered conditionally on covariates when the source propensity $e(X)$ is bounded away from zero and one.

The full-data and sampling primitives are fixed by the following population and array conditions.

Assumption 1. `[ass:full-data-support]` (Full-data support) *The full-data law satisfies*

$$P_n^F \{S \in \{0, 1\}, D(0), D(1) \in \{0, 1\}, Y(0), Y(1) \in [0, 1]\} = 1.$$

This support restriction is specific to this analysis: it places the ratio parameter on a bounded outcome scale and makes the induced complier contrast live in a fixed compact interval.

Assumption 2. `[ass:population-presence]` (Population presence) *For each $s \in \{0, 1\}$,*

$$P_n^F(S = s) > 0.$$

Population presence is the standard two-population support condition, ensuring that the source and target conditional laws are well defined [Chen and Huang, 2025].

Assumption 3. `[ass:two-sample-array]` (Two-sample array) *For a transported triangular array on the measurable covariate space $\mathcal{X}$, let P_S denote the index- n observed source law induced on $O_i^S = (X_i, Z_i, D_i, Y_i)$, and let P_T denote the index- n target covariate law on X_j^T . The two sample factors satisfy:*

- *(Source factor.) For every n , P_S is a probability measure.*
- *(Target factor.) For every n , P_T is a probability measure.*
- *(Sample-size ratio.) $0 < c$ and $N_n/n \rightarrow c$.*

The observed two-sample triangular-array sampling world treats the source observations O_i^S and target covariates X_j^T as the two empirical inputs, with source law P_S , target covariate law P_T , target sample size N_n , and positive sample-size limit c . This is the standard independent two-sample triangular-array sampling condition used in transported noncompliance designs [Chen and Huang, 2025].

The next block gives the source instrumental-variables structure. These restrictions follow the usual LATE logic for an encouragement design, adapted to the transported two-population notation of the present problem [Imbens and Angrist, 1994, Angrist et al., 1996].

Assumption 4. `[ass:instrument-overlap]` (Instrument overlap) *The instrument-overlap constant satisfies $0 < \varepsilon < 1/2$, and the source propensity score satisfies*

$$\varepsilon \leq e(X) \leq 1 - \varepsilon$$

P_S^X -almost surely.

Instrument overlap is the standard instrument positivity condition, here indexed by the constant ε , and it keeps inverse-propensity contrasts uniformly controlled [Chen and Huang, 2025].

Assumption 5. `[ass:source-observation]` (Source observation law) *Conditional on $S = 1$, the observed source law satisfies:*

- *(Assignment propensity.) $P(Z = 1 \mid X, S = 1) = e(X)$.*
- *(Treatment receipt.) $D = D(Z)$.*
- *(Observed outcome.) $Y = Y(D)$.*

The source observation law is the standard assignment and consistency condition linking the observed source experiment to receipt and outcome potential outcomes [Chen and Huang, 2025].

Assumption 6. `[ass:iv-randomization]` (Instrument randomization) *The instrument satisfies*

$$Z \perp (Y(0), Y(1), D(0), D(1)) \mid (X, S = 1).$$

Instrument randomization is the standard conditional instrument exchangeability condition; within source covariate strata, the encouragement is as good as randomized for the relevant potential outcomes [Chen and Huang, 2025].

Assumption 7. `[ass:iv-exclusion]` (Instrument exclusion) *For every $z, s \in \{0, 1\}$,*

$$Y^Z(z) = Y(D(z))$$

P_n^F -almost surely conditional on $S = s$.

The exclusion restriction is the standard condition that the assignment potential outcome $Y^Z(z)$ operates through treatment receipt [Chen and Huang, 2025].

Assumption 8. `[ass:iv-monotonicity]` (Instrument monotonicity) *For every $s \in \{0, 1\}$,*

$$D(1) \geq D(0)$$

P_n^F -almost surely conditional on $S = s$.

Monotonicity is the standard no-defiers condition, allowing the receipt contrast to be interpreted as a complier share in each population state [Chen and Huang, 2025].

Transport enters through an outcome-contrast restriction, an integrated first-stage restriction, and covariate-support inclusion. The outcome condition is conditional in X , whereas the receipt condition is imposed at the target average; this distinction is the transport structure used by the identification result below.

Assumption 9. `[ass:outcome-transport]` (Outcome-contrast transport) *For P_T -almost every x ,*

$$\mathbb{E}_{P_n^F}[Y^Z(1) - Y^Z(0) \mid X = x, S = 0] = \mathbb{E}_{P_n^F}[Y^Z(1) - Y^Z(0) \mid X = x, S = 1].$$

Outcome-contrast transport is the standard conditional transport of the assignment-outcome contrast across the two population states [Chen and Huang, 2025].

Assumption 10. `[ass:receipt-transport]` (Receipt-contrast transport) *The target-population average of the conditional treatment-receipt contrast agrees across the two population states:*

$$\mathbb{E}_{X|S=0}[\mathbb{E}_{P_n^F}[D(1) - D(0) \mid X, S = 0]] = \mathbb{E}_{X|S=0}[\mathbb{E}_{P_n^F}[D(1) - D(0) \mid X, S = 1]].$$

Receipt-contrast transport equates the target-law average compliance margin across populations while allowing covariate-specific receipt contrasts to differ [Chen and Huang, 2025]. This condition is the denominator link that represents the target complier share by the source conditional first stage averaged over P_T .

Assumption 11. `[ass:target-complier-positivity]` (Target complier positivity) *The target population has positive complier mass:*

$$P_n^F(D(1) > D(0) \mid S = 0) > 0.$$

Target complier positivity is the standard positive target complier share condition, so the transported complier ratio is well defined [Chen and Huang, 2025].

Assumption 12. [ass:transport-domination] (Transport domination) *The target covariate law is absolutely continuous with respect to the source covariate marginal on $\mathcal{X}$:*

$$P_T \ll P_S^X.$$

Transport domination is the standard covariate-support inclusion condition for generalizing source contrasts to the target covariate distribution [Chen and Huang, 2025]. It yields the transport and overlap geometry through the source marginal P_S^X , the density-ratio weight $w(X) = dP_T/dP_S^X(X)$, and the source propensity.

The next definition collects the identification functionals built from this geometry: the source conditional contrasts, their transported means, and the effective-strength index.

Definition 2. [synth_6] (Transported contrasts and effective strength) *Let $w(X) = dP_T/dP_S^X(X)$ be a nonnegative version of the transport density ratio. The source conditional assignment-outcome and receipt contrasts are*

$$\Delta_Y(x) = \mathbb{E}_{P_n^F}[Y^Z(1) - Y^Z(0) \mid X = x, S = 1], \quad \Delta_D(x) = \mathbb{E}_{P_n^F}[D(1) - D(0) \mid X = x, S = 1].$$

Their transported means and the transport-weight dispersion are

$$\mu_{Y,n} = \mathbb{E}_{P_S^X}[w(X) \Delta_Y(X)], \quad \mu_n = \mathbb{E}_{P_S^X}[w(X) \Delta_D(X)], \quad \kappa_n = \mathbb{E}_{P_S^X}[w(X)^2],$$

and the effective identification strength at index n is $t_n = n\mu_n^2/\kappa_n$.

The remaining restrictions define the weak-first-stage geometry. They separate the size of the transport weights from the transported first stage, producing a single effective-strength index for honest expected length.

Assumption 13. [ass:weight-envelope] (Weight envelope) *The transport weight satisfies*

$$0 \leq w(X) \leq 2k_n$$

P_S -almost surely.

The weight envelope is specific to this analysis: the sequence k_n controls the largest admissible density-ratio values while permitting the amount of covariate shift to grow with n .

Assumption 14. [ass:weight-second-moment] (Weight second moment) *The second moment of the transport weight satisfies*

$$\kappa_n \leq k_n.$$

The second-moment bound is specific to this analysis and records the Kish-type dispersion scale $\kappa_n = \mathbb{E}_S[w(X)^2]$ that governs the variance of transported inverse-propensity averages.

Assumption 15. [ass:degrading-array] (Degrading array rates) *The array satisfies*

$$k_n \rightarrow \infty, \quad k_n = o(n^{1/2}), \quad \mu_n \rightarrow 0.$$

These rate restrictions are specific to this analysis: the envelope scale diverges at a sub-root- n rate, and the transported first-stage mean μ_n enters the weak-identification regime. Together with the dispersion scale, they define the effective-strength scalar $t_n = n\mu_n^2/\kappa_n$ used in the risk comparison.

Assumption 16. `[ass:finite-cell-source]` (Uniform finite cells) *The source covariate marginal P_S^X is supported on exactly k_n measurable cells, assigns mass $1/k_n$ to each cell, and satisfies*

$$e(X) = 1/2$$

P_S -almost surely.

The uniform finite-cell condition is specific to this analysis. It creates the sample-only cell-learning submodel used for feasible procedures, with equal source cell masses and a randomized source encouragement probability.

The full model and the finite-cell submodel collect these assumptions at each row of the triangular array.

Definition 3. `[def:model-class]` (Transported LATE model class $\mathcal{P}_n$) *The transported LATE model class at index n is*

$$\mathcal{P}_n = \{(P_n^F, e) : (P_n^F, e) \text{ satisfies the following conditions at index } n\}.$$

The required conditions are:

- *(Full-data support.)* [assumption 1.](#)
- *(Population presence.)* [assumption 2.](#)
- *(Two-sample array.)* [assumption 3.](#)
- *(Instrument overlap.)* [assumption 4.](#)
- *(Source observation.)* [assumption 5.](#)
- *(Instrument randomization.)* [assumption 6.](#)
- *(Instrument exclusion.)* [assumption 7.](#)
- *(Instrument monotonicity.)* [assumption 8.](#)
- *(Outcome transport.)* [assumption 9.](#)
- *(Receipt transport.)* [assumption 10.](#)
- *(Target complier positivity.)* [assumption 11.](#)
- *(Transport domination.)* [assumption 12.](#)
- *(Weight envelope.)* [assumption 13.](#)
- *(Weight second moment.)* [assumption 14.](#)
- *(Degrading array.)* [assumption 15.](#)

The transported LATE model class $\mathcal{P}_n$ is the main domain for the oracle minimax statements. It combines the source IV restrictions with the transport and dispersion conditions that make the ratio target comparable across rows.

The finite-cell class is the transported model with one added row restriction:

$$\mathcal{N}_n = \{P \in \mathcal{P}_n : P \text{ satisfies } \text{assumption 16}\}.$$

Thus the IV, transport, overlap, envelope, second-moment, and degrading-array conditions are inherited from $\mathcal{P}_n$. The added content is the uniform source-cell geometry in [assumption 16](#); the anchored checklist form of $\mathcal{N}_n$ is deferred to [section 6](#).

Before defining risk, we record the oracle inputs and the two score radii used later. Oracle procedures receive the source sample, the target covariate sample, the transport weight, and the source propensity.

Definition 4. [\[synth_2\]](#) (Score-inversion critical radius L_α) *We define the score-inversion critical radius by $L_\alpha = \{8/(\alpha\varepsilon^2)\}^{1/2}$, where $\alpha \in (0, 1)$ is the noncoverage level and ε is the source instrument-overlap constant.*

The constant in [definition 4](#) combines the noncoverage level α with instrument overlap. It will determine the radius of the score-inversion set in the oracle construction.

The coverage requirements below are stated for the following causal estimand.

Definition 5. [\[synth_7\]](#) (Target complier effect θ_T) *The target complier effect is the ratio*

$$\theta_T = \frac{\mathbb{E}_{P_n^F}[\{Y(1) - Y(0)\} \mathbf{1}\{D(1) = 1, D(0) = 0\} \mid S = 0]}{P_n^F\{D(1) = 1, D(0) = 0 \mid S = 0\}},$$

the mean effect of treatment receipt among target-population compliers. Under target complier positivity the denominator is positive and $\theta_T = \mathbb{E}_{P_n^F}[Y(1) - Y(0) \mid D(1) = 1, D(0) = 0, S = 0]$; when the denominator is zero, the ratio is taken to be zero.

Definition 6. [\[def:oracle-honesty\]](#) (Oracle honest class $\mathfrak{C}^{\text{or}}$) *The oracle honest class is the class of oracle procedures $(C_n)_{n \geq 1}$ satisfying $0 < \alpha < 1$ and the following conditions. For every n , C_n maps each input*

$$\left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w, e \right), \quad w : \mathcal{X} \rightarrow \mathbb{R}, \quad w(x) \geq 0, \quad e : \mathcal{X} \rightarrow \mathbb{R},$$

to a subset of $\Theta = [-1, 1]$. For every n and every measurable such pair (w, e) , the sample-by-parameter graph

$$\left\{ \left(((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}), \vartheta \right) : \vartheta \in C_n \left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w, e \right) \right\}$$

is measurable. For every n , every $P \in \mathcal{P}_n$, and every pair w, w' of measurable nonnegative versions of dP_T/dP_S^X , the evaluations using the row- n propensity score agree for the two-sample law almost surely:

$$C_n \left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w, e \right) = C_n \left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w', e \right).$$

Finally,

$$\liminf_{n \rightarrow \infty} \begin{cases} \inf_{P \in \mathcal{P}_n} P(\theta_T \in C_n), & \mathcal{P}_n \neq \emptyset, \\ 1, & \mathcal{P}_n = \emptyset \end{cases} \geq 1 - \alpha,$$

where in the probability term C_n is evaluated at the canonical row- n density ratio dP_T/dP_S^X and the row- n propensity score.

The procedure and minimax-risk world treats each confidence set C_n as a random subset of the compact parameter space $\Theta = [-1, 1]$. Oracle honesty, written $\mathfrak{C}^{\text{or}}$, requires uniform asymptotic coverage over $\mathcal{P}_n$ while allowing the procedure to use the canonical nonnegative weight input w and the propensity.

Definition 7. `[def:frontier-risk]` (Frontier risk $R((C_n), t_0)$) *For a confidence-set sequence $(C_n)_{n \geq 1}$ and threshold t_0 , the limiting worst-case expected length above effective strength t_0 is*

$$R((C_n), t_0) = \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)].$$

The risk $R((C_n), t_0)$ measures limiting worst-case Lebesgue length $\lambda(C_n)$ over laws whose effective strength exceeds the positive threshold t_0 . This conditioning by t_n is the paper's way of comparing honest procedures at a fixed level of transported identification strength.

Definition 8. `[def:oracle-value]` (Oracle value $V^*(t_0)$) *For any fixed effective-strength threshold t_0 with $0 < t_0$, the oracle minimax expected-length value is*

$$V^*(t_0) = \inf_{(C_n) \in \mathfrak{C}^{\text{or}}} R((C_n), t_0).$$

The value $V^*(t_0)$ is the oracle benchmark over honest procedures with known transport weights and propensities.

Definition 9. `[def:finite-cell-oracle-value]` (Finite-cell oracle value $V_{\mathcal{N}}^*(t_0)$) *The finite-cell oracle minimax expected-length value at threshold t_0 is*

$$V_{\mathcal{N}}^*(t_0) = \inf_{(C_n) \in \mathfrak{C}^{\text{or}}} \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)].$$

The finite-cell oracle value $V_{\mathcal{N}}^*(t_0)$ applies the same oracle procedure class to the uniform finite-cell submodel. It gives the comparison point for feasible cell-based weight learning.

Definition 10. `[def:finite-cell-feasible-honesty]` (Feasible finite-cell class $\mathfrak{C}^{\text{cell}}$) *The feasible finite-cell honest class is*

$$\mathfrak{C}^{\text{cell}} = \left\{ (C_n)_{n \geq 1} : C_n = C_n \left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n} \right) \subseteq \Theta, \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{N}_n} P(\theta_T \in C_n) \geq 1 - \alpha \right\}.$$

The feasible class $\mathfrak{C}^{\text{cell}}$ uses the source observations and target covariates as the procedure inputs in the uniform finite-cell design. Its coverage requirement remains uniform over $\mathcal{N}_n$.

The identification result connects the transported functionals of [definition 2](#) to the target complier effect of [definition 5](#): the inverse-propensity score recovers the source conditional contrasts Δ_Y and Δ_D from observed source moments, transport moves $\mu_{Y,n}$ and μ_n to target complier functionals, and the Wald ratio equals θ_T .

Theorem 1. `[prop:compact-causal-range]` (Compact causal range) *Fix n , and let P be a transported array with sample-size and envelope schedules N_n and k_n , constants c and ε , and model class $\mathcal{P}_n$ as in [definition 3](#). Suppose that:*

- **(Model membership.)** $P \in \mathcal{P}_n$ at the index n .
- **(Source contrasts.)** The assignment outcome contrast and the receipt contrast for the source population satisfy the usual measurability and integrability conditions under the source covariate marginal P_S^X .

Then Δ_Y and Δ_D represent the observed source outcome and receipt contrasts: for every measurable covariate set A ,

$$\int_{\{O: X \in A\}} H(O)Y dP_S(O) = \int_A \Delta_Y(x) dP_S^X(x), \quad \int_{\{O: X \in A\}} H(O)D dP_S(O) = \int_A \Delta_D(x) dP_S^X(x).$$

Moreover, the transported reduced-form outcome mean satisfies

$$\mu_{Y,n} = \mathbb{E}_{P_n^F}[\{Y(1) - Y(0)\}\mathbf{1}\{D(1) = 1, D(0) = 0\} \mid S = 0],$$

the transported first-stage mean satisfies

$$\mu_n = P_n^F\{D(1) = 1, D(0) = 0 \mid S = 0\},$$

and the Wald ratio equals the target complier-conditional contrast,

$$\frac{\mu_{Y,n}}{\mu_n} = \theta_T.$$

Finally, $\theta_T \in \Theta = [-1, 1]$.

Theorem 1 establishes the identification link used throughout the paper: the weighted source reduced form equals the target complier-weighted outcome contrast, the weighted source first stage equals the target complier share, and their ratio is the target complier-conditional contrast. The bounded outcome support then places the estimand in Θ , making expected length on a fixed compact parameter space the relevant risk criterion.

For later fixed-geometry arguments, we also record the notation for the centered least-favorable law and the two distance measures used to compare local alternatives.

Definition 11. [synth_5] (Centered fixed-geometry law $Q_{n,0}^g$) We write $Q_{n,0}^g$ for the $h = 0$ member of the fixed-geometry least-favorable family: for the fixed admissible geometry $\mathbf{g} = (P_S^X, w, e)_{n \geq 1}$, conditional on $X = x$ the compliance types $(D(0), D(1)) = (0, 1), (1, 1), (0, 0)$ have probabilities $p_n(x), (1 - p_n(x))/2, (1 - p_n(x))/2$, $Y(0) = 0$, $Y(1)$ is Bernoulli(1/2) for every compliance type, the same conditional full-data law is used in both populations, source encouragement is drawn with propensity $e(X)$, and the notation denotes the resulting joint observed-data law together with the fixed oracle inputs.

The centered law in [definition 11](#) is the baseline member of the fixed-geometry least-favorable family used in the lower-bound construction.

Definition 12. [synth_3] (Chi-square separation of the fixed-geometry submodel) For the fixed-geometry laws $Q_{n,h}^g$ and $Q_{n,0}^g$, we write $\chi^2(Q_{n,h}^g \| Q_{n,0}^g)$ for the chi-square divergence

$$\chi^2(Q_{n,h}^g \| Q_{n,0}^g) = \int \left(\frac{dQ_{n,h}^g}{dQ_{n,0}^g} - 1 \right)^2 dQ_{n,0}^g,$$

whenever $Q_{n,h}^g \ll Q_{n,0}^g$, with value $+\infty$ otherwise.

Definition 13. `[synth_4]` (Total variation separation of least-favorable laws) *We write $\|Q_{n,h}^{\mathfrak{g}} - Q_{n,0}^{\mathfrak{g}}\|_{\text{TV}}$ for the total variation distance between the index- n joint observed-data laws with fixed oracle inputs generated by the geometry $\mathfrak{g}$ at local complier-outcome tilt h and at the center $h = 0$, namely $\sup_A |Q_{n,h}^{\mathfrak{g}}(A) - Q_{n,0}^{\mathfrak{g}}(A)|$ over measurable events A in the corresponding source-sample, target-covariate-sample, and oracle-input space.*

The separations in [definitions 12](#) and [13](#) calibrate the statistical distance between fixed-geometry local alternatives. They provide the notation needed to express how lower-bound perturbations move the transported complier contrast while remaining close in observed-data law.

3 Oracle Frontier and Score Inversion

The minimax comparison now turns from identification to expected length. The relevant scalar is the effective strength $t_n = n\mu_n^2/\kappa_n$: the numerator records the transported first-stage signal and the denominator records the Kish dispersion of the transport weights. Weak-IV robust confidence sets are naturally built by inverting a reduced-form score restriction, following the same principle behind Anderson–Rubin and Fieller inference [[Anderson and Rubin, 1949](#), [Fieller, 1954](#), [Staiger and Stock, 1997](#), [Andrews et al., 2019](#)]. In the transported problem, the oracle score weights each source observation by the known density ratio and uses the known source propensity.

Definition 14. `[def:inversion-handle]` (Score-inversion set C_n) *For each $\vartheta \in \Theta$, define the source score*

$$w(X) \left\{ \frac{Z}{e(X)} - \frac{1-Z}{1-e(X)} \right\} (Y - \vartheta D).$$

Define the empirical second moment of the oracle weights by

$$\hat{\kappa}_n = \frac{1}{n} \sum_{i=1}^n w(X_i)^2.$$

With oracle-known w and e , the score-inversion confidence set is

$$C_n = \left\{ \vartheta \in \Theta : \left| \frac{1}{n} \sum_{i=1}^n w(X_i) \left\{ \frac{Z_i}{e(X_i)} - \frac{1-Z_i}{1-e(X_i)} \right\} (Y_i - \vartheta D_i) \right| \leq L_\alpha \sqrt{\frac{\hat{\kappa}_n}{n}} \right\}.$$

For oracle-known w and e , this score is a source-sample statistic.

The set in [definition 14](#) contains exactly those candidate ratios whose transported reduced-form residual is small on the scale $L_\alpha \sqrt{\hat{\kappa}_n}/n$. The radius uses the empirical second moment of the oracle weights, so covariate shift enters the score through the same dispersion term that appears in the effective-strength index. Because the construction is restricted to Θ , its length is measured on the compact causal range established in [theorem 1](#).

The first result gives the converse side of the oracle benchmark. It compares all oracle honest procedures in $\mathfrak{C}^{\text{or}}$ under the full transported class and under the uniform finite-cell submodel.

Theorem 2. `[thm:oracle-converse]` (Oracle frontier converse) *Fix $c > 0$, $\varepsilon \in (0, 1/2)$, and $\alpha \in (0, 1)$. There exist constants $c_0 > 0$ and $t_c > 0$, depending on c , ε , and α , such that the following holds.*

- **(Array size.)** The target sample sizes satisfy $N_n/n \rightarrow c$.
- **(Cell scale.)** The sequence k_n is positive, satisfies $k_n \rightarrow \infty$, and satisfies $k_n/\sqrt{n} \rightarrow 0$.
- **(Measurable finite carriers.)** For each n , the covariate space contains k_n distinct measurable singleton carrier cells.
- **(Transported-IV class.)** Every array in the transported model class $\mathcal{P}_n$ of [definition 3](#) satisfies [assumptions 1 to 15](#).
- **(Finite-cell submodel.)** Every array in the finite-cell class $\mathcal{N}_n$ of [definition 22](#) satisfies [assumption 16](#).

Then, for every $t_0 > 0$, the oracle values of [definitions 8 and 9](#) obey

$$c_0 \min\{1, t_0^{-1/2}\} \leq V^\star(t_0)$$

and

$$c_0 \min\{1, t_0^{-1/2}\} \leq V_{\mathcal{N}}^\star(t_0).$$

Moreover, for every $t_0 \in (0, t_c]$,

$$c_0 \leq V^\star(t_0).$$

The same constants c_0 and t_c apply uniformly over all decompositions of the effective strength t_0 into the transported first-stage mean μ_n and the weight dispersion κ_n .

[Theorem 2](#) establishes that honest expected length is bounded below at the order $\min\{1, t_0^{-1/2}\}$. The constants c_0 and t_c depend on the sampling ratio, overlap, and noncoverage level, while the final uniformity clause indexes the lower bound by the combined effective strength uniformly over decompositions into first-stage weakness and weight dispersion. The finite-cell inequality records the same frontier lower bound inside the uniform-cell submodel, which is the model used for feasible weight learning in the next section.

The matching construction is the score inversion from [definition 14](#). Its role is parallel to Anderson–Rubin confidence sets for weak instruments: each hypothesized ratio turns the problem into a reduced-form moment condition, and inversion of that moment yields an honest set even when the denominator is small [[Anderson and Rubin, 1949](#), [Dufour, 1997](#), [Stock et al., 2002](#), [Moreira, 2003](#), [Andrews et al., 2006](#), [Mikusheva, 2010](#), [Andrews et al., 2019](#)].

The attainment theorem uses an ambient nonemptiness hypothesis to ensure that the comparison domain contains an admissible geometry. The score-inversion sequence itself is global: when it is evaluated at a law in $\mathcal{P}_n$, it uses that law’s canonical density ratio and row- n propensity.

Theorem 3. [\[thm:oracle-score-inversion-attainment\]](#) (Oracle score attainment) *Fix sequences N_n and k_n , constants c, ε, α , the transported-IV model class $\mathcal{P}_n$ in [definition 3](#), and the score-inversion construction in [definition 14](#). Assume:*

- **(Sampling ratio.)** $c > 0$ and $N_n/n \rightarrow c$.
- **(Overlap and noncoverage.)** $0 < \varepsilon < 1/2$ and $0 < \alpha < 1$.
- **(Envelope scale.)** $k_n > 0$ for every n , $k_n \rightarrow \infty$, and $k_n/\sqrt{n} \rightarrow 0$.

- **(Nonempty admissible carrier.)** There is an admissible deterministic geometry $\mathfrak{g} \in \mathcal{G}$ satisfying [definition 15](#) at scale k_n and overlap constant ε .
- **(Uniform model restrictions.)** Every $P \in \mathcal{P}_n$ satisfies the two-sample condition in [assumption 3](#), the overlap condition in [assumption 4](#), the weight envelope in [assumption 13](#), the weight second-moment bound in [assumption 14](#), and the degrading-array condition in [assumption 15](#).

Let

$$L_\alpha = \left(\frac{8}{\alpha \varepsilon^2} \right)^{1/2}.$$

For each n , define

$$H_i = \frac{Z_i}{e(X_i)} - \frac{1 - Z_i}{1 - e(X_i)}, \quad \hat{A}_n = \frac{1}{n} \sum_{i=1}^n w(X_i) H_i Y_i, \quad \hat{B}_n = \frac{1}{n} \sum_{i=1}^n w(X_i) H_i D_i,$$

and

$$\hat{\kappa}_n = \frac{1}{n} \sum_{i=1}^n w(X_i)^2, \quad C_n = \left\{ \vartheta \in \Theta : \left| \hat{A}_n - \vartheta \hat{B}_n \right| \leq L_\alpha \sqrt{\frac{\hat{\kappa}_n}{n}} \right\}.$$

Then the score-inversion sequence (C_n) belongs to the oracle honest class $\mathfrak{C}^{\text{or}}$ of [definition 6](#). Moreover, with

$$C_0 = \max \left\{ 2, 4L_\alpha + \frac{8}{\varepsilon^2} \right\},$$

for every $t_0 > 0$, the frontier risk from [definition 7](#) satisfies

$$R((C_n), t_0) \leq C_0 \min\{1, t_0^{-1/2}\}.$$

Consequently the oracle value in [definition 8](#) satisfies the same attainable frontier order over fixed positive effective-strength thresholds.

[Theorem 3](#) gives the upper-bound side. The statistic $\hat{A}_n - \vartheta \hat{B}_n$ is the transported sample analogue of the reduced form for $Y - \vartheta D$, and the confidence set collects the values of ϑ whose score lies within the oracle radius. The constant C_0 is universal over the stated class once α and ε are fixed through L_α .

The carrier clause supplies the nonempty comparison domain used to state attainability. The transported score statistic in [theorem 3](#) uses the w and e supplied by each evaluated law.

Combining [theorems 2](#) and [3](#), the oracle minimax expected length is characterized, up to constants, by

$$\min\{1, t_0^{-1/2}\}.$$

For small effective strength, the compact parameter space yields a constant-length frontier. As t_0 grows, the length contracts at the inverse-square-root rate in the effective strength. The rate therefore treats a weak transported first stage and high transport dispersion symmetrically through $n\mu_n^2/\kappa_n$, which is the quantity carried forward to the feasible finite-cell procedures.

4 Fixed Geometry and Feasible Cell Weight Learning

The oracle comparison in [theorems 2](#) and [3](#) treats the transport weight and source propensity as inputs to the procedure. This section separates two questions that are useful for interpreting that benchmark. First, it holds the transport geometry fixed and records the local least-favorable family that delivers the same inverse-square-root dependence on effective strength. Second, it shows how the weight-dispersion scale can be learned from target covariates in finite-cell experiments while preserving the expected-length order.

A fixed geometry records the source covariate law, the target-to-source weight, and the source propensity as a deterministic triangular array. The following definition isolates the admissible arrays used throughout the section.

Definition 15. `[def:admissible-geometry-class]` (Admissible geometry class $\mathcal{G}$) $\mathcal{G}$ is the class of deterministic arrays $\mathbf{g} = (P_S^X, w, e)_{n \geq 1}$ satisfying the following conditions:

- *(Source law.)* P_S^X is a probability law on $\mathcal{X}$.
- *(Instrument overlap.)* $e : \mathcal{X} \rightarrow [\varepsilon, 1 - \varepsilon]$.
- *(Transport weights.)* $w : \mathcal{X} \rightarrow [0, 2k_n]$, $\mathbb{E}_S[w(X)] = 1$, and

$$\kappa_n = \mathbb{E}_S[w(X)^2] \leq k_n.$$

- *(Target law.)* For every measurable $A \subseteq \mathcal{X}$,

$$P_T(A) = \mathbb{E}_S[w(X)\mathbf{1}\{X \in A\}].$$

The class $\mathcal{G}$ fixes the $\mathbf{g}$ component of the transported experiment while retaining the same overlap, envelope, normalization, and second-moment controls used in the oracle frontier. The target law is generated from the source covariate law by the density ratio, so the geometry keeps the covariate-shift component explicit and leaves the causal response law to vary within the model.

Definition 16. `[def:fixed-geometry-slice]` (Fixed-geometry slice $\mathcal{P}_n(\mathbf{g})$) For $\mathbf{g} \in \mathcal{G}$,

$$\mathcal{P}_n(\mathbf{g}) = \{P \in \mathcal{P}_n : (P_S^X, w, e) \text{ equals the index-} n \text{ component of } \mathbf{g}\}.$$

The slice $\mathcal{P}_n(\mathbf{g})$ is the subproblem in which the causal and outcome laws vary subject to the transported LATE restrictions, while the source covariate marginal, target density ratio, and propensity are held fixed.

Definition 17. `[def:fixed-geometry-honesty]` (Fixed-geometry oracle honesty $\mathfrak{C}^{\text{or}}(\mathbf{g})$) For $\mathbf{g} \in \mathcal{G}$ as in [definition 15](#), $\mathfrak{C}^{\text{or}}(\mathbf{g})$ is the class of sequences $(C_n)_{n \geq 1}$ such that

$$C_n = C_n\left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w, e\right) \subseteq \Theta$$

and

$$\liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{P}_n(\mathbf{g})} P(\theta_T \in C_n) \geq 1 - \alpha.$$

The class $\mathfrak{C}^{\text{or}}(\mathfrak{g})$ imposes honesty over the fixed-geometry slice. Procedures may use the fixed oracle inputs, and the coverage requirement remains uniform over the causal response laws compatible with that geometry.

Definition 18. [def:fixed-geometry-value] (Fixed-geometry value $V_{\mathfrak{g}}^*(t_0)$) For $\mathfrak{g} \in \mathcal{G}$ as in [definition 15](#), the fixed-geometry minimax expected-length value at threshold t_0 is

$$V_{\mathfrak{g}}^*(t_0) = \inf_{(C_n) \in \mathfrak{C}^{\text{or}}(\mathfrak{g})} \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}_n(\mathfrak{g}): t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)].$$

The value $V_{\mathfrak{g}}^*(t_0)$ measures the same expected-length criterion as [definition 8](#), now conditional on a fixed admissible geometry. This formulation makes clear that the effective-strength comparison is a statement about the response-law variation remaining after the covariate-shift array has been fixed.

The lower-bound construction varies a single complier-outcome tilt while preserving the fixed geometry. It uses the same target-first-stage calibration as the oracle converse, with a local radius chosen so that the observed-data laws remain close in total variation.

Definition 19. [def:least-favorable-witness] (Least-favorable witness $Q_{n,h}^{\mathfrak{g}}$) Fix $\mathfrak{g} = (P_S^X, w, e)_{n \geq 1} \in \mathcal{G}$ and $t_0 > 0$. Set

$$\mu_n = \left(\frac{t_0 \kappa_n}{n} \right)^{1/2}, \quad p_n(x) = \frac{\mu_n w(x)}{\kappa_n}, \quad \rho = \frac{1 - \alpha}{8}, \quad H(t_0) = \min\{1/4, \rho t_0^{-1/2}\}.$$

For each $|h| \leq H(t_0)$, conditional on $X = x$, assign compliance types by

$$\Pr\{(D(0), D(1)) = (0, 1) \mid X = x\} = p_n(x),$$

$$\Pr\{(D(0), D(1)) = (1, 1) \mid X = x\} = \Pr\{(D(0), D(1)) = (0, 0) \mid X = x\} = \frac{1 - p_n(x)}{2}.$$

Set $Y(0) = 0$. For compliers, let $Y(1)$ be Bernoulli with mean $1/2 + h$; for always-takers and never-takers, let $Y(1)$ be Bernoulli with mean $1/2$. Use this same conditional full-data law in both populations, draw source encouragement with propensity $e(X)$, and write $Q_{n,h}^{\mathfrak{g}}$ for the resulting joint observed-data law together with the fixed oracle inputs.

In [definition 19](#), the complier probability $p_n(x)$ is proportional to the transport weight, so the transported first stage is exactly calibrated to the threshold t_0 . The parameter ρ fixes the Le Cam separation scale, and the tilt h moves the target complier contrast while preserving the same source and target covariate geometry.

The notation for the centered law and the two distances is fixed in [definitions 11 to 13](#). The fixed-geometry lower bound also uses the separation calculation recorded in [definition 23](#): a tilt of order $t_0^{-1/2}$ moves the transported complier contrast while the observed-data laws remain close in chi-square and total variation.

The next theorem gives the fixed-geometry frontier for arbitrary admissible deterministic geometries. It combines the lower-bound calibration in [definitions 19 and 23](#) with the oracle score construction from [theorem 3](#).

Theorem 4. [thm:fixed-geometry-frontier] (Fixed-geometry frontier) Let $c > 0$, $0 < \varepsilon < 1/2$, and $0 < \alpha < 1$. Assume the following conditions hold:

- **(Target-sample ratio.)** The target sample sizes satisfy $N_n/n \rightarrow c$.
- **(Envelope scale.)** The sequence k_n is positive for every n , satisfies $k_n \rightarrow \infty$, and obeys $k_n/\sqrt{n} \rightarrow 0$.
- **(Geometry.)** The deterministic geometry array $\mathbf{g}$ satisfies [definition 15](#) with scale k_n and overlap constant ε .
- **(Fixed-geometry slices.)** For every n and every transported array whose n th law lies in the fixed-geometry slice $\mathcal{P}_n(\mathbf{g})$ of [definition 16](#), the conditions in [assumptions 3, 4 and 13 to 15](#) hold.

For every $t_0 > 0$, define

$$L_\alpha = \left(\frac{8}{\alpha \varepsilon^2} \right)^{1/2}, \quad c_\alpha = \frac{3(1-\alpha)^2}{16}, \quad C_\alpha = \max \left\{ 2, 4L_\alpha + \frac{8}{\varepsilon^2} \right\}.$$

Then the fixed-geometry minimax value of [definition 18](#) satisfies

$$c_\alpha \min\{1, t_0^{-1/2}\} \leq V_{\mathbf{g}}^*(t_0) \leq C_\alpha \min\{1, t_0^{-1/2}\}.$$

The constants c_α and C_α are uniform over admissible deterministic geometries $\mathbf{g}$.

[Theorem 4](#) establishes that fixing the covariate-shift geometry leaves the same expected-length order as the full oracle comparison: $\min\{1, t_0^{-1/2}\}$. The constants are uniform over the admissible geometry class, so the effective-strength index absorbs both first-stage weakness and weight dispersion across the fixed-geometry slices.

A useful specialization arises when the target and source covariate laws coincide. In that case the dispersion term collapses to one, and the transported effective strength reduces to the familiar sample-size-scaled first-stage square.

Theorem 5. [prop:no-shift-reduction] (Single-population frontier reduction) *On a measurable covariate space $\mathcal{X}$, let $\mathbf{g}$ be a deterministic geometry array satisfying [definition 15](#) with scale k_n and overlap constant ε . Suppose that*

- **(Sampling ratio.)** $c > 0$ and $N_n/n \rightarrow c$.
- **(Overlap and noncoverage.)** $0 < \varepsilon < 1/2$ and $0 < \alpha < 1$.
- **(Cell scale.)** $k_n > 0$ for every n , $k_n \rightarrow \infty$, and $k_n/\sqrt{n} \rightarrow 0$.
- **(Unit transport weight.)** For every n , $w_n(X) = 1$ $P_{S,n}^X$ -almost surely under $\mathbf{g}$.

Then, for every n ,

$$\kappa_n = 1 \quad \text{and} \quad P_{T,n} = P_{S,n}^X.$$

Moreover, for every transported array P and every n , if $P \in \mathcal{P}_n(\mathbf{g})$ in the sense of [definition 16](#), then its effective strength satisfies

$$t_n = n\mu_n^2.$$

Finally, for every $t_0 > 0$, let

$$L_\alpha = \sqrt{\frac{8}{\alpha \varepsilon^2}}, \quad c_\alpha = \frac{3(1-\alpha)^2}{16}, \quad C_\alpha = \max \left\{ 2, 4L_\alpha + \frac{8}{\varepsilon^2} \right\}.$$

The fixed-geometry minimax value $V_g^*(t_0)$ from [definition 18](#) obeys

$$c_\alpha \min\{1, t_0^{-1/2}\} \leq V_g^*(t_0) \leq C_\alpha \min\{1, t_0^{-1/2}\}.$$

[Theorem 5](#) records the unit-weight specialization of [theorem 4](#). With unit transport weight, the target law equals the source covariate law, $\kappa_n = 1$, and the rate is indexed by $n\mu_n^2$. The same compact-parameter expected-length order therefore recovers the usual weak-first-stage scaling within the transported notation.

We now turn from oracle-known weights to sample-only finite-cell score inversion. In the uniform finite-cell model of [definition 22](#), the target sample identifies the target cell frequencies, while the source sample estimates the cellwise outcome and receipt contrasts. This is the finite-cell analogue of inverse-probability weighting in semiparametric transport and missing-data problems [[Hirano et al., 2003](#), [Robins et al., 1994](#), [van der Laan and Rubin, 2006](#), [Chernozhukov et al., 2018](#)].

Theorem 6. `[thm:finite-cell-unknown-weight-attainment]` (Finite-cell sample-only score-inversion attainment) *Let N_n and k_n be positive integer sequences, let $c, \varepsilon, \alpha \in \mathbb{R}$, and suppose the following conditions hold.*

- **(Sampling ratio.)** $c > 0$ and $N_n/n \rightarrow c$.
- **(Overlap and coverage levels.)** $0 < \varepsilon < 1/2$ and $0 < \alpha < 1$.
- **(Cell growth.)** $k_n > 0$ for every n , $k_n \rightarrow \infty$, and $k_n/\sqrt{n} \rightarrow 0$.
- **(Ambient carrier.)** For every n , the measurable covariate space contains k_n distinct measurable singleton cells.
- **(Finite-cell rows.)** For every row n and every transported array $P \in \mathcal{N}_n$, the row satisfies the two-sample condition in [assumption 3](#), the uniform finite-cell source condition in [assumption 16](#), and the degrading-array condition in [assumption 15](#), with $\mathcal{N}_n$ as in [definition 22](#).

Put

$$B_c = 32(1 + c^{-1}), \quad L_{\alpha,c} = \left(\frac{2B_c}{\alpha}\right)^{1/2}, \quad C_0 = \max\{2, 4\sqrt{2}L_{\alpha,c} + 8B_c\}.$$

Then $C_0 > 0$, and there exists a sample-only finite-cell confidence-set sequence $C = (C_n)$ in the feasible class $\mathfrak{C}^{\text{cell}}$ of [definition 10](#) such that, for every n and every $P \in \mathcal{N}_n$, C_n agrees P -almost surely with the finite-cell score-inversion rule using critical constant $L_{\alpha,c}$. Equivalently, on the finite-cell experiment, this rule is given by

$$H_i = 2(2Z_i - 1), \quad \hat{p}_{x,n} = \frac{1}{N_n} \sum_{j=1}^{N_n} \mathbf{1}\{X_j^T = x\},$$

and, for $G \in \{Y, D\}$,

$$\hat{m}_{G,n}(x) = \frac{k_n}{n} \sum_{i=1}^n \mathbf{1}\{X_i = x\} H_i G_i, \quad \widehat{M}_{G,n} = \sum_{x=1}^{k_n} \hat{p}_{x,n} \hat{m}_{G,n}(x).$$

For $N_n \geq 2$, define

$$\hat{\kappa}_{U,n} = \frac{k_n}{N_n(N_n - 1)} \sum_{j \neq \ell} \mathbf{1}\{X_j^T = X_\ell^T\}, \quad \hat{K}_n = 1 + \hat{\kappa}_{U,n},$$

and set

$$C_n = \left\{ \vartheta \in \Theta : |\widehat{M}_{Y,n} - \vartheta \widehat{M}_{D,n}| \leq L_{\alpha,c} \sqrt{\hat{K}_n/n} \right\},$$

with $C_n = \Theta$ at indices for which $N_n < 2$. For every $t_0 > 0$,

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)] \leq C_0 \min\{1, t_0^{-1/2}\}.$$

Moreover, for every $t_0 > 0$, the feasible finite-cell honest minimax risk satisfies

$$\frac{3(1-\alpha)^2}{16} \min\{1, t_0^{-1/2}\} \leq \inf_{\tilde{C} \in \mathfrak{C}^{\text{cell}}} \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(\tilde{C}_n)].$$

Theorem 6 constructs a feasible confidence set from the source observations and target covariates. The empirical target frequencies $\hat{p}_{x,n}$ estimate the transported mixing weights, the cellwise contrasts $\hat{m}_{G,n}(x)$ estimate the source reduced-form and first-stage contrasts, and $\hat{\kappa}_{U,n}$ is a target-sample U-statistic for the weight-dispersion scale. The resulting inversion set has the same $\min\{1, t_0^{-1/2}\}$ expected-length order as the oracle benchmark on the uniform finite-cell class, with the lower bound showing that this order is minimax sharp within $\mathfrak{C}^{\text{cell}}$.

For the uniform-cell benchmark, the inputs are the source records (X, Z, D, Y) , the target covariates X^T , the known cell partition, and the known propensity $1/2$. Tabulate target frequencies, compute $H_i = 2(2Z_i - 1)$ and the source cell contrasts, aggregate the two transported moments, estimate dispersion with the target collision statistic, and invert the displayed affine score over Θ . The regular-cell version follows the same steps while taking the known $q_{x,n}$ and $e_{x,n}$ as additional inputs. These constructions are minimax attainability benchmarks under the stated growth conditions; their declared inputs fix the cell structure and propensities used by the rules.

The uniform-cell construction uses known equal source cell masses and the propensity $e(X) = 1/2$. A regular finite-cell variant allows unequal but controlled source cell probabilities and a known cell-varying propensity. The source-cell probabilities $q_{x,n}$ are bounded between constants c_- and c_+ times $1/k_n$, which preserves the same dispersion order while accommodating regular discrete designs.

Definition 20. [def:regular-finite-cell-class] (Regular finite-cell class $\mathcal{N}_n^{\text{reg}}$) For fixed constants $0 < c_- \leq 1 \leq c_+ < \infty$, define

$$\mathcal{N}_n^{\text{reg}} = \left\{ P \in \mathcal{P}_n : \begin{array}{l} P_S^X \text{ is supported on exactly } k_n \text{ measurable cells,} \\ q_{x,n} = P_S^X(X \text{ lies in cell } x), \\ c_-/k_n \leq q_{x,n} \leq c_+/k_n \text{ for every } x \end{array} \right\}.$$

The regular class $\mathcal{N}_n^{\text{reg}}$ keeps the finite support structure while allowing cell masses to vary within fixed multiplicative bounds. This is the discrete counterpart of overlap and bounded-density conditions commonly used in transport and weighting analyses [Chen and Huang, 2025, Rudolph and Laan, 2016, Zeng et al., 2024].

Definition 21. [def:regular-cell-feasible-honesty] (Regular-cell feasible class $\mathfrak{C}^{\text{reg}}$) For a noncoverage level $0 < \alpha < 1$, define $\mathfrak{C}^{\text{reg}}$ as the class of regular-cell procedures $(C_n)_{n \geq 1}$ with the following properties. For each n , C_n is a Θ -valued random set whose inputs are $(O_i^S)_{i=1}^n$, $(X_j^T)_{j=1}^{N_n}$, the realized regular-cell support, and the known cell-indexed vectors $(q_{x,n})_{x=1}^{k_n}$ and $(e_{x,n})_{x=1}^{k_n}$. For every realized regular-cell support and every pair of real k_n -vectors $(q_{x,n})_{x=1}^{k_n}$ and $(e_{x,n})_{x=1}^{k_n}$, the sample-by- θ graph of C_n , restricted to source and target samples on that support, is measurable. These procedures satisfy

$$1 - \alpha \leq \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{N}_n^{\text{reg}}} P(\theta_T \in C_n),$$

with the rowwise infimum evaluated as 1 when $\mathcal{N}_n^{\text{reg}}$ is empty.

The feasible class $\mathfrak{C}^{\text{reg}}$ formalizes procedures that know the regular source-cell probabilities and the cell-indexed propensity, but learn the target cell frequencies from the target sample. The measurability clause ensures that the inversion rule is a well-defined random set on each realized finite carrier.

Theorem 7. [thm:regular-cell-unknown-weight-attainment] (Regular-cell weight attainment) Fix $c, \varepsilon, \alpha, c_-, c_+ \in \mathbb{R}$ and sequences N_n and k_n . Suppose that

- **(Sample-size ratio.)** $0 < c$ and $N_n/n \rightarrow c$.
- **(Overlap and coverage levels.)** $0 < \varepsilon < 1/2$ and $0 < \alpha < 1$.
- **(Regular cell constants.)** $0 < c_- \leq 1$ and $1 \leq c_+$.
- **(Cell growth.)** $k_n > 0$ for every n , $k_n \rightarrow \infty$, and $k_n/\sqrt{n} \rightarrow 0$.
- **(Measurable carrier.)** For every n , the covariate carrier contains k_n injected measurable singleton cells.
- **(Model restrictions.)** Every array $P \in \mathcal{N}_n^{\text{reg}}$ from [definition 20](#) with constants c, ε, c_-, c_+ satisfies [assumption 3](#), [assumption 4](#), and [assumption 15](#).

On $\mathcal{N}_n^{\text{reg}}$, let $q_{x,n} = P_S^X(X = x)$ and define

$$H_i = \frac{Z_i}{e(X_i)} - \frac{1 - Z_i}{1 - e(X_i)}.$$

For $G \in \{Y, D\}$, set

$$\hat{p}_{x,n} = \frac{1}{N_n} \sum_{j=1}^{N_n} \mathbf{1}\{X_j^T = x\}, \quad \hat{m}_{G,n}(x) = \frac{1}{nq_{x,n}} \sum_{i=1}^n \mathbf{1}\{X_i = x\} H_i G_i,$$

and

$$\widehat{M}_{G,n} = \sum_{x=1}^{k_n} \hat{p}_{x,n} \hat{m}_{G,n}(x).$$

For $N_n \geq 2$, set

$$\hat{\kappa}_{U,n} = \frac{1}{N_n(N_n - 1)} \sum_{j \neq \ell} \frac{\mathbf{1}\{X_j^T = X_\ell^T\}}{q_{X_j^T,n}}, \quad \hat{K}_n = 1 + \hat{\kappa}_{U,n},$$

and put $C_n^{\text{reg}} = \Theta$ when $N_n < 2$. Let

$$B_{\varepsilon,c} = 8(\varepsilon^{-2} + c^{-1}).$$

For every fixed $L \geq \{2B_{\varepsilon,c}/\alpha\}^{1/2}$, the sequence

$$C_n^{\text{reg}} = \left\{ \vartheta \in \Theta : |\widehat{M}_{Y,n} - \vartheta \widehat{M}_{D,n}| \leq L \sqrt{\widehat{K}_n/n} \right\}$$

for $N_n \geq 2$, together with the preceding convention for $N_n < 2$, belongs to $\mathfrak{C}^{\text{reg}}$ from [definition 21](#) and is constructed from the known regular source-cell probabilities, the known propensity, and the source and target samples. Moreover, for every fixed $t_0 > 0$,

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n^{\text{reg}}: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n^{\text{reg}})] \leq \max\{2, 4\sqrt{2}L + 8B_{\varepsilon,c}\} \min\{1, t_0^{-1/2}\}.$$

The corresponding minimax risk over $\mathfrak{C}^{\text{reg}}$ is bounded below by

$$\frac{3(1-\alpha)^2}{16} \min\{1, t_0^{-1/2}\}.$$

Thus, with known regular source-cell probabilities and a known cell-varying propensity satisfying the stated overlap, the feasible regular-cell construction attains the oracle frontier order on $\mathcal{N}_n^{\text{reg}}$.

[Theorem 7](#) extends finite-cell score inversion to regular source-cell masses and known cell-varying propensities. The statistic uses inverse-propensity source contrasts, target empirical cell frequencies, and a regular-cell U-statistic dispersion proxy. Under the stated sampling, overlap, and cell-growth conditions, the construction is honest over $\mathfrak{C}^{\text{reg}}$ and attains the same expected-length order as the fixed-geometry and uniform finite-cell procedures.

Together, [theorems 4, 6 and 7](#) show how the oracle frontier carries over from fixed weights to sample-learned finite-cell weights. The common rate $\min\{1, t_0^{-1/2}\}$ is indexed by the effective strength $n\mu_n^2/\kappa_n$, with κ_n interpreted either as an oracle dispersion measure or as a target-sample-estimated finite-cell collision scale.

5 Related Literature and Discussion

The paper belongs first to the LATE and principal-stratification literature. The source experiment uses the encouragement-design logic of [Bloom \[1984\]](#), [Imbens and Angrist \[1994\]](#), and [Angrist et al. \[1996\]](#): conditional randomization, exclusion, and monotonicity turn the source reduced form and first stage into a complier contrast. Principal stratification supplies the population-level interpretation of compliance types and target complier effects [[Balke and Pearl, 1997](#), [Imbens and Rubin, 1997](#), [Frangakis and Rubin, 2002](#), [Imbens and Rosenbaum, 2005](#), [Heckman and Vytlacil, 2005](#)]. In the present setting, [theorem 1](#) identifies the transported reduced form and transported first stage through weighted source contrasts and places the target complier contrast inside the compact parameter space $\Theta = [-1, 1]$.

The weak-identification component follows the robust ratio-inference tradition. Anderson–Rubin and Fieller procedures invert reduced-form score restrictions, giving ratio inference through the score restriction itself [[Anderson and Rubin, 1949](#), [Fieller, 1954](#)]. Subsequent work characterizes the consequences of weak instruments for distributional approximation, confidence

sets, and expected length [Gleser and Hwang, 1987, Dufour, 1997, Bound et al., 1995, Staiger and Stock, 1997, Stock et al., 2002, Moreira, 2003, Andrews et al., 2006, Mikusheva, 2010, Andrews et al., 2019, Bertanha and Moreira, 2020, Ma, 2023, Smucler et al., 2025]. The oracle score construction in definition 14 is the transported analogue of this logic: it tests candidate values by centering the weighted reduced form after subtracting the candidate times the weighted first stage. Theorems 2 and 3 then characterize the minimax expected-length order for the transported ratio as $\min\{1, t_0^{-1/2}\}$.

Transportability and generalizability enter through the target-to-source covariate weighting structure. Work on transporting and generalizing treatment effects emphasizes support, covariate reweighting, and target-population interpretation [Joseph Hotz et al., 2005, Pearl and Bareinboim, 2011, Bareinboim and Pearl, 2013, 2016, Cole and Stuart, 2010, Stuart et al., 2011, Tipton, 2013, Dahabreh et al., 2020b,a, Nguyen et al., 2017]. Here the transported-IV restrictions in definition 3 combine those support and contrast conditions with an IV compliance structure. The effective-strength scalar $t_n = n\mu_n^2/\kappa_n$ summarizes the transported first-stage mean μ_n and Kish dispersion κ_n , so the expected-length comparison uses a single scale for weak compliance and covariate reweighting.

The feasible procedures connect to semiparametric weighting and modern transported noncompliance methods. Inverse-probability weighting, augmented estimation, and sample-splitting ideas motivate estimating transported nuisance components while retaining valid target-population interpretation [Hirano et al., 2003, Robins et al., 1994, van der Laan and Rubin, 2006, Chernozhukov et al., 2018].

Transported noncompliance. Chen and Huang [2025], cited as an arXiv working paper, identify and estimate transported CACE in a two-sample generalizability design with source noncompliance and target population information; their inferential object is transported-CACE identification and estimation. Rudolph and Laan [2016], a journal article, develop robust estimators for encouragement-design effects transported across sites. Ross et al. [2026], also a journal article, study trial transport when participation changes adherence, with transported trial effects as the inferential target. The present paper adds the honest minimax expected-length frontier for a compact transported-CACE ratio under a triangular-array weak first stage and Kish weight dispersion, together with finite-cell procedures that attain the oracle order.

Weak-IV and two-sample IV inference. Ma [2023], cited as an arXiv working paper, gives identification-robust LATE inference with high-dimensional covariates. Smucler et al. [2025], also cited as an arXiv working paper, studies the properties of LATE confidence sets obtained by score-test inversion. Choi et al. [2018] and Choi and Shen [2019] instead give weak-instrument-robust inference for linear IV coefficients assembled from two samples. The present paper’s source experiment and target covariate sample define a covariate-transport problem, and its positive scope is an honest frontier indexed by $n\mu_n^2/\kappa_n$ with finite-cell target-weight learning.

Relative to transported-ATE efficiency and minimax benchmarks, the parameter here is a compact transported complier ratio whose denominator may shrink along the triangular array. Under the stated transported-IV, bounded-outcome, overlap, and dispersion conditions, the frontier is indexed by the ratio strength $n\mu_n^2/\kappa_n$, whereas regular transported-ATE benchmarks use root- n target-mean scales.

Relative to classical weak-IV and two-sample IV inference, the present setting adds target-to-source covariate weighting and finite-cell target-law learning. Under the stated finite-cell and

regular-cell conditions, the expected-length order records first-stage weakness and the dispersion induced by transporting the covariate distribution through the single scale $n\mu_n^2/\kappa_n$.

Taken together, the results give a finite expected-length characterization for transported complier ratios in the stated transported-IV model classes, with the stated overlap, transport, positivity, envelope, dispersion, sample-size, and cell-growth conditions. The compact causal range in [theorem 1](#) keeps length measured on Θ , while [theorem 4](#) shows that the fixed-geometry minimax value is indexed by the effective-strength threshold t_0 . This synthesis places transported noncompliance on the same weak-ratio inference scale as classical weak-IV analysis, with κ_n recording the additional variation induced by moving from the source covariate law to the target covariate law.

6 Appendix: Proofs, Auxiliary Arguments, and Verification Note

The appendix gives the mathematical arguments in dependency order. It begins with the compact causal-range calculation, then proves the fixed-geometry frontier and its unit-weight specialization, then gives the oracle lower and upper bounds, and finally treats the regular and uniform finite-cell constructions. Throughout, confidence sets take values in the compact parameter space Θ , and expected length is the Lebesgue length criterion from [definition 7](#).

The finite-cell submodel is $\mathcal{P}_n$ plus the uniform finite-cell source condition in [assumption 16](#). For reference, the anchored formal checklist is:

Definition 22. `[def:finite-cell-class]` (Finite-cell class $\mathcal{N}_n$) *The finite-cell submodel at index n is*

$$\mathcal{N}_n = \{P \in \mathcal{P}_n : P \text{ satisfies the following conditions at index } n\}.$$

The required conditions are:

- *(Full-data support.)* [assumption 1](#).
- *(Population presence.)* [assumption 2](#).
- *(Two-sample array.)* [assumption 3](#).
- *(Instrument overlap.)* [assumption 4](#).
- *(Source observation.)* [assumption 5](#).
- *(Instrument randomization.)* [assumption 6](#).
- *(Instrument exclusion.)* [assumption 7](#).
- *(Instrument monotonicity.)* [assumption 8](#).
- *(Outcome transport.)* [assumption 9](#).
- *(Receipt transport.)* [assumption 10](#).
- *(Target complier positivity.)* [assumption 11](#).
- *(Transport domination.)* [assumption 12](#).

- (*Weight envelope.*) [assumption 13](#).
- (*Weight second moment.*) [assumption 14](#).
- (*Degrading array.*) [assumption 15](#).
- (*Finite-cell source.*) [assumption 16](#).

The fixed-geometry lower bounds use the centered law and distance notation in [definitions 11 to 13](#). The following anchored statement records the separation calculation for the least-favorable laws generated by [definition 19](#).

Definition 23. `[def:geometry-handle]` (Geometry handle $Q_{n,h}^g$) *Fix an admissible deterministic array $\mathbf{g} = (P_S^X, w, e)_{n \geq 1}$. For $t_0 > 0$, set*

$$\mu_n = \left(\frac{t_0 \kappa_n}{n} \right)^{1/2}, \quad p_n(x) = \frac{\mu_n w(x)}{\kappa_n}.$$

For $0 < \alpha < 1$, let $\rho = (1 - \alpha)/8$. As the complier outcome mean varies over $h \in \mathbb{R}$ with

$$|h| \leq \min\{1/4, \rho t_0^{-1/2}\},$$

using the chi-square divergence and total variation distance defined in [definitions 12 and 13](#), the corresponding least-favorable laws satisfy

$$1 + \chi^2(Q_{n,h}^g \| Q_{n,0}^g) \leq \exp(8t_0 h^2), \quad \|Q_{n,h}^g - Q_{n,0}^g\|_{\text{TV}} \leq 2\rho.$$

The proof of [theorem 1](#) expands the inverse-propensity score in [definition 1](#) conditional on X . Conditional instrument randomization and the source observation law turn the source score moments into the conditional assignment contrasts $\Delta_Y(x)$ and $\Delta_D(x)$. Exclusion and monotonicity express those contrasts through complier potential outcomes, while [assumptions 9, 10 and 12](#) move the reduced form and first stage to the target covariate law. The bounded outcome support in [assumption 1](#) gives the compact range $[-1, 1]$, so the transported Wald ratio is interpreted as the target complier-conditional contrast in [theorem 1](#).

The lower-bound arguments for [theorems 2 and 4](#) use the least-favorable family in [definition 19](#). That construction sets the complier probability proportional to the transport weight, which calibrates the transported first stage to the selected effective-strength threshold. Varying the complier outcome tilt changes the target complier contrast by a controlled amount, while [definition 23](#) bounds the chi-square and total-variation separations between the tilted law and the centered law. Le Cam’s testing argument then converts closeness of the observed-data laws into a lower bound on the expected length of any honest confidence set. The single-population specialization in [theorem 5](#) follows by setting $w = 1$, which gives $\kappa_n = 1$ and the effective-strength index $n\mu_n^2$.

For the oracle score-inversion result, [definition 14](#) is an Anderson–Rubin style inversion of the transported residual moment, in the tradition of [Anderson and Rubin \[1949\]](#) and [Fieller \[1954\]](#). At the true value θ_T , the centered score has variance controlled by the oracle weight second moment and the overlap constant. The radius $L_\alpha \sqrt{\widehat{\kappa}_n/n}$ combines that variance control with a uniform probability bound, yielding the honesty assertion in [theorem 3](#). The expected-length calculation uses the same score inequality together with the effective-strength lower restriction $t_n \geq t_0$, producing the order $\min\{1, t_0^{-1/2}\}$.

The finite-cell arguments show that sample-learned weights preserve the oracle rate. For [theorem 6](#), the target empirical frequencies estimate target cell probabilities, the source cellwise averages estimate reduced-form and first-stage contrasts, and the U-statistic $\hat{\kappa}_{U,n}$ estimates the collision scale associated with target-to-source weight dispersion. The sub-root cell-growth condition ensures that the additional sampling variation from cell learning remains on the same scale as the oracle score radius. The regular-cell extension in [theorem 7](#) repeats this control with known source-cell masses $q_{x,n}$ and known cell-varying propensity, using the regularity constants c_- and c_+ to keep the cellwise variance terms uniformly bounded.

Verification note. The top-level theorem statements and proof obligations corresponding to [theorems 1 to 7](#) are checked in Lean 4. The auxiliary notation objects [definitions 1, 4 and 11 to 13](#) organize the manuscript notation used by those checked declarations, and [definition 23](#) records the checked fixed-geometry separation calculation used in the lower-bound proofs.

Proof of [theorem 1](#). Throughout, write $F = (S, X, D(0), D(1), Y(0), Y(1))$ for the full-data record, and let

$P_{\text{assign},S}$ = the joint law of (F, Z) in the source encouragement experiment.

Model membership in $\mathcal{P}_n$, as specified in [definition 3](#), supplies through [assumption 5](#) that $P_{\text{assign},S}$ is a probability law whose full-data marginal is the source full-data law and that the observed source law is its image under the observation map g :

$$\text{law of } F \text{ under } P_{\text{assign},S} = P_n^F(\cdot \mid S = 1), \quad g(F, Z) = (X, Z, D(Z), Y(D(Z))),$$

$$P_S = \text{law of } g(F, Z) \text{ under } P_{\text{assign},S}.$$

The map g is measurable: its first two coordinates are coordinate projections, and its last two are two-branch selections between projections indexed by the measurable sets $\{Z = 1\}$ and $\{D(Z) = 1\}$. We also abbreviate

$$\mu = P_n^F(\cdot \mid S = 0), \quad C = \{D(1) = 1, D(0) = 0\},$$

and we fix, alongside the source contrasts Δ_Y, Δ_D of [definition 2](#), measurable versions of the corresponding target conditional contrasts

$$\begin{aligned} \Delta_Y^T(x) &= \mathbb{E}_{P_n^F}[Y^Z(1) - Y^Z(0) \mid X = x, S = 0], \\ \Delta_D^T(x) &= \mathbb{E}_{P_n^F}[D(1) - D(0) \mid X = x, S = 0]. \end{aligned}$$

Finally, H is the score of [definition 1](#); in the two branches of the binary instrument it takes the values

$$H(O) = \begin{cases} 1/e(X), & Z = 1, \\ -1/\{1 - e(X)\}, & Z = 0. \end{cases}$$

1. **Source marginals, overlap on the observed law, and the score bound.** The first coordinate of g is X , so composing the two displays above shows that the source covariate marginal is also the covariate marginal of the source full-data law:

$$P_S^X = \text{law of } X \text{ under } P_n^F(\cdot \mid S = 1).$$

Since P_S^X is by construction the X -marginal of P_S , the overlap statement of [assumption 4](#) transfers verbatim to the observed source law and, by the last display, to the source full-data law:

$$\varepsilon \leq e(X) \leq 1 - \varepsilon \quad P_S\text{-a.s. and } P_n^F(\cdot \mid S = 1)\text{-a.s.}$$

The score H is measurable, being a two-branch selection between $1/e$ and $-1/(1-e)$ along the measurable set $\{Z = 1\}$, and the two branches are bounded as follows: on $\{Z = 1\}$, $|H(O)| = 1/e(X) \leq 1/\varepsilon$ because $e(X) \geq \varepsilon > 0$; on $\{Z = 0\}$, $|H(O)| = 1/\{1 - e(X)\} \leq 1/\varepsilon$ because $1 - e(X) \geq \varepsilon > 0$. Hence

$$|H(O)| \leq 1/\varepsilon \quad P_S\text{-a.s.}$$

By [assumption 1](#) the potential outcomes lie in $[0, 1]$, and $P_n^F(\cdot \mid S = 1) \ll P_n^F$; pushing this through g , whose outcome coordinate is one of $Y(0), Y(1)$, gives $Y \in [0, 1]$ P_S -almost surely, while $D = D(Z) \in \{0, 1\}$ by construction. Therefore

$$|H(O)Y| \leq 1/\varepsilon, \quad |H(O)D| \leq 1/\varepsilon \quad P_S\text{-a.s.},$$

and since P_S is a probability law both $H(O)Y$ and $H(O)D$ are P_S -integrable.

2. **The instrument slice of the assigned source law.** For each $z \in \{0, 1\}$ put

$$\pi_z(F) = z e(X) + (1 - z)\{1 - e(X)\}.$$

Conditional randomization in [assumption 6](#), combined with the assignment-propensity clause $P(Z = 1 \mid X, S = 1) = e(X)$ of [assumption 5](#), says exactly that for every measurable full-data set B ,

$$P_{\text{assign}, S}\{(F, Z) : F \in B, Z = z\} = \int_B \pi_z(F) dP_n^F(\cdot \mid S = 1).$$

The overlap bound of the previous step gives $0 < \varepsilon \leq \pi_z \leq 1 - \varepsilon < 1$ $P_n^F(\cdot \mid S = 1)$ -almost surely, so π_z is a nonnegative integrable density and $P_n^F(\cdot \mid S = 1)$ is a probability law; the restricted image measure on the left is finite. A finite measure whose mass on every measurable set equals the integral of a fixed nonnegative integrable density is that weighted measure, so the displayed set identity upgrades to the measure identity

$$\text{image under } F \text{ of } P_{\text{assign}, S} \text{ restricted to } \{Z = z\} = \pi_z \cdot P_n^F(\cdot \mid S = 1).$$

3. **The inverse-propensity contrast identity for full-data integrands.** Let F_0, F_1 be measurable full-data functions, integrable under $P_n^F(\cdot \mid S = 1)$, and suppose that

$$(F, Z) \mapsto \frac{\mathbf{1}\{Z = 1\}}{e(X)} F_1 - \frac{\mathbf{1}\{Z = 0\}}{1 - e(X)} F_0$$

is $P_{\text{assign}, S}$ -integrable. Fix a measurable $A \subseteq \mathcal{X}$. Splitting the domain $\{X \in A\}$ into its two instrument slices, the integrand reduces on each slice to a function of F alone, so the measure identity of the previous step applies after the change of variables $(F, Z) \mapsto F$:

$$\int_{\{X \in A, Z=1\}} \frac{F_1}{e(X)} dP_{\text{assign}, S} = \int_{\{X \in A\}} \frac{1}{e(X)} e(X) F_1 dP_n^F(\cdot \mid S = 1)$$

$$\begin{aligned}
&= \int_{\{X \in A\}} F_1 dP_n^F(\cdot \mid S = 1), \\
\int_{\{X \in A, Z=0\}} \frac{F_0}{1 - e(X)} dP_{\text{assign}, S} &= \int_{\{X \in A\}} \frac{1}{1 - e(X)} \{1 - e(X)\} F_0 dP_n^F(\cdot \mid S = 1) \\
&= \int_{\{X \in A\}} F_0 dP_n^F(\cdot \mid S = 1),
\end{aligned}$$

where the cancellations are legitimate because $e(X) \geq \varepsilon > 0$ and $1 - e(X) \geq \varepsilon > 0$ almost surely. Subtracting the second line from the first,

$$\int_{\{X \in A\}} \left[\frac{\mathbf{1}\{Z = 1\}}{e(X)} F_1 - \frac{\mathbf{1}\{Z = 0\}}{1 - e(X)} F_0 \right] dP_{\text{assign}, S} = \int_{\{X \in A\}} (F_1 - F_0) dP_n^F(\cdot \mid S = 1).$$

4. **The observed source representation of Δ_Y .** The chosen version Δ_Y is measurable, and we claim that for every measurable $A \subseteq \mathcal{X}$,

$$\int_{\{O: X \in A\}} H(O)Y dP_S(O) = \int_A \Delta_Y(x) dP_S^X(x),$$

which is the first asserted representation. Changing variables through the representation of P_S as the image of $P_{\text{assign}, S}$ under g , and using that g sends (F, Z) to $(X, Z, D(Z), Y(D(Z)))$, so that $H(O)Y$ pulls back to the displayed inverse-propensity integrand with $F_z = Y(D(z))$,

$$\int_{\{O: X \in A\}} H(O)Y dP_S(O) = \int_{\{X \in A\}} \left[\frac{\mathbf{1}\{Z = 1\}}{e(X)} Y(D(1)) - \frac{\mathbf{1}\{Z = 0\}}{1 - e(X)} Y(D(0)) \right] dP_{\text{assign}, S}.$$

The two integrands $Y(D(0)), Y(D(1))$ are measurable and, by [assumption 1](#), bounded by one, hence $P_n^F(\cdot \mid S = 1)$ -integrable; and the assigned-source integrand above is the pull-back through g of $H(O)Y$, whose P_S -integrability was established in the first step. The identity of the previous step therefore applies and gives

$$\int_{\{O: X \in A\}} H(O)Y dP_S(O) = \int_{\{X \in A\}} \{Y(D(1)) - Y(D(0))\} dP_n^F(\cdot \mid S = 1).$$

By [assumption 7](#) at $s = 1$ we may replace $Y(D(z))$ by $Y^Z(z)$ up to a null set, and then the defining conditional-expectation property of Δ_Y in [definition 2](#), together with the identification of P_S^X with the source full-data covariate marginal from the first step, gives

$$\int_{\{X \in A\}} \{Y^Z(1) - Y^Z(0)\} dP_n^F(\cdot \mid S = 1) = \int_A \Delta_Y(x) dP_S^X(x),$$

which completes the claim.

5. **The observed source representation of Δ_D .** The chosen version Δ_D is measurable, and the same three moves, with the real-valued receipt indicators $F_z = D(z) \in \{0, 1\}$ in place of $Y(D(z))$ and with no exclusion step needed, give for every measurable $A \subseteq \mathcal{X}$

$$\int_{\{O: X \in A\}} H(O)D dP_S(O) = \int_{\{X \in A\}} \left[\frac{\mathbf{1}\{Z = 1\}}{e(X)} D(1) - \frac{\mathbf{1}\{Z = 0\}}{1 - e(X)} D(0) \right] dP_{\text{assign}, S}$$

$$= \int_{\{X \in A\}} \{D(1) - D(0)\} dP_n^F(\cdot \mid S = 1),$$

the first equality by the change of variables through g (which sends the observed receipt coordinate to $D(Z)$) and the second by the contrast identity, whose integrability hypotheses hold because $D(0), D(1)$ are bounded by one and because the assigned-source integrand is the pull-back of the P_S -integrable $H(O)D$. The defining conditional-expectation property of Δ_D in [definition 2](#), again read against P_S^X , turns the right-hand side into $\int_A \Delta_D(x) dP_S^X(x)$, so

$$\int_{\{O: X \in A\}} H(O)D dP_S(O) = \int_A \Delta_D(x) dP_S^X(x),$$

which is the second asserted representation.

6. **The transported reduced-form outcome mean.** By [definition 2](#), $\mu_{Y,n} = \int w(x) \Delta_Y(x) dP_S^X(x)$. Since $P_T \ll P_S^X$ by [assumption 12](#) and w is a version of dP_T/dP_S^X , the change of measure

$$\int w(x) \Delta_Y(x) dP_S^X(x) = \int \Delta_Y(x) dP_T(x)$$

holds. [Assumption 9](#) states $\Delta_Y^T = \Delta_Y$ P_T -almost surely, and P_T is the covariate marginal of μ , so applying the defining property of Δ_Y^T with $A = \mathcal{X}$,

$$\int \Delta_Y(x) dP_T(x) = \int \Delta_Y^T(x) dP_T(x) = \int \{Y^Z(1) - Y^Z(0)\} d\mu.$$

[Assumption 7](#) at $s = 0$ rewrites the integrand as $Y(D(1)) - Y(D(0))$. [Assumption 8](#) at $s = 0$ makes the pair $(D(0), D(1)) = (1, 0)$ μ -null, so off a μ -null set exactly one of the three remaining cases holds, and in each of them the integrand coincides with the complier-weighted contrast:

$$Y(D(1)) - Y(D(0)) = \begin{cases} Y(0) - Y(0) = 0 = \{Y(1) - Y(0)\} \mathbf{1}_C, & (D(0), D(1)) = (0, 0), \\ Y(1) - Y(1) = 0 = \{Y(1) - Y(0)\} \mathbf{1}_C, & (D(0), D(1)) = (1, 1), \\ Y(1) - Y(0) = \{Y(1) - Y(0)\} \mathbf{1}_C, & (D(0), D(1)) = (0, 1). \end{cases}$$

Hence

$$\mu_{Y,n} = \int \{Y(1) - Y(0)\} \mathbf{1}_{\{D(1) = 1, D(0) = 0\}} dP_n^F(\cdot \mid S = 0).$$

7. **The transported first-stage mean.** By [definition 2](#), $\mu_n = \int w(x) \Delta_D(x) dP_S^X(x)$. The relevant integrands are bounded: $0 \leq w \leq 2k_n$ by [assumption 13](#) and $0 \leq \Delta_D \leq 1$ because Δ_D is the conditional mean of $D(1) - D(0) \in \{0, 1\}$ under [assumption 8](#), so $|w \Delta_D| \leq 2k_n$ and all integrals below are absolutely convergent. Exactly as in the previous step, transport domination and $w = dP_T/dP_S^X$ give

$$\int w(x) \Delta_D(x) dP_S^X(x) = \int \Delta_D(x) dP_T(x),$$

and [assumption 10](#), which equates the two P_T -averages of the conditional receipt contrasts rather than the contrasts themselves, gives

$$\int \Delta_D(x) dP_T(x) = \int \Delta_D^T(x) dP_T(x) = \int \{D(1) - D(0)\} d\mu,$$

the second equality being the defining property of Δ_D^T with $A = \mathcal{X}$. [Assumption 8](#) at $s = 0$ again removes $(D(0), D(1)) = (1, 0)$, and in the three remaining cases

$$D(1) - D(0) = \begin{cases} 0 = \mathbf{1}_C, & (D(0), D(1)) = (0, 0), \\ 0 = \mathbf{1}_C, & (D(0), D(1)) = (1, 1), \\ 1 = \mathbf{1}_C, & (D(0), D(1)) = (0, 1), \end{cases}$$

so $D(1) - D(0) = \mathbf{1}_C$ μ -almost surely. Integrating the indicator,

$$\mu_n = \int \mathbf{1}_C d\mu = \mu(C) = P_n^F\{D(1) = 1, D(0) = 0 \mid S = 0\}.$$

8. **The Wald ratio.** By [definition 5](#),

$$\theta_T = \frac{\int \{Y(1) - Y(0)\} \mathbf{1}_C d\mu}{\mu(C)}.$$

The previous two steps identify the numerator with $\mu_{Y,n}$ and the denominator with μ_n , so

$$\frac{\mu_{Y,n}}{\mu_n} = \theta_T.$$

9. **The forced range.** [Assumption 1](#) makes P_n^F a probability law, and [assumption 2](#) gives $P_n^F(S = 0) > 0$, so the conditional law μ is a well-defined probability law and $\mu \ll P_n^F$. Consequently the bounded-support clause of [assumption 1](#) transfers to μ , and subtracting the two bounds $Y(0), Y(1) \in [0, 1]$ yields

$$-1 \leq Y(1) - Y(0) \leq 1 \quad \mu\text{-a.s.}$$

The set C is measurable, so $\mathbf{1}_C \in \{0, 1\}$; multiplying the last display by $\mathbf{1}_C$ gives $-\mathbf{1}_C \leq \{Y(1) - Y(0)\} \mathbf{1}_C \leq \mathbf{1}_C$ μ -almost surely, all three functions being bounded and hence μ -integrable. Integral monotonicity then gives

$$-\mu(C) = -\int \mathbf{1}_C d\mu \leq \int \{Y(1) - Y(0)\} \mathbf{1}_C d\mu \leq \int \mathbf{1}_C d\mu = \mu(C).$$

[Assumption 11](#) gives $\mu(C) > 0$; dividing the display by this positive number and comparing with the expression for θ_T in the previous step yields

$$\theta_T = \frac{\int \{Y(1) - Y(0)\} \mathbf{1}_C d\mu}{\mu(C)} \in [-1, 1] = \Theta.$$

□

Proof of [theorem 4](#). Throughout, $\Theta = [-1, 1]$, $\lambda(A)$ is the Lebesgue length of $A \cap \Theta$, and for a row law P we abbreviate the functionals of [definition 2](#),

$$\mu_{Y,n} = \mathbb{E}_{P_S^X}[w(X)\Delta_Y(X)], \quad \mu_n = \mathbb{E}_{P_S^X}[w(X)\Delta_D(X)], \quad \kappa_n = \mathbb{E}_{P_S^X}[w(X)^2], \quad t_n = \frac{n\mu_n^2}{\kappa_n},$$

writing $t_n(P)$ and $\theta_T(P)$ when the law has to be displayed. Fix $t_0 > 0$.

Step 0 (the slice is eventually inhabited). The admissible geometry $\mathfrak{g}$ of [definition 15](#), together with $N_n/n \rightarrow c > 0$, $k_n > 0$, $k_n \rightarrow \infty$, and $k_n/\sqrt{n} \rightarrow 0$, supplies an explicit row law at every late index: for all sufficiently large n ,

$$\exists P \in \mathcal{P}_n(\mathfrak{g}) \quad \text{with} \quad t_n(P) \geq t_0.$$

In particular $\mathcal{P}_n(\mathfrak{g}) \neq \emptyset$ at those indices, so the coverage row appearing in [definition 17](#) is an infimum over a nonempty set and the strength-restricted supremum in [definition 18](#) is over a nonempty set.

Part A: the lower bound.

Step 1 (the least-favorable family). Following [definition 19](#), set the Le Cam separation scale ρ and the outcome-tilt radius $H(t_0)$,

$$\rho = \frac{1-\alpha}{8}, \quad H(t_0) = \min \left\{ \frac{1}{4}, \rho t_0^{-1/2} \right\}.$$

Since $0 < \alpha < 1$ and $t_0 > 0$ we have $\rho > 0$ and $H(t_0) > 0$, and by construction $H(t_0) \leq 1/4$. For all sufficiently large n the construction of [definition 19](#) supplies a family $\{Q_{n,h}^{\mathfrak{g}} : |h| \leq H(t_0)\}$ with

$$Q_{n,h}^{\mathfrak{g}} \in \mathcal{P}_n(\mathfrak{g}), \quad t_n(Q_{n,h}^{\mathfrak{g}}) = t_0, \quad \theta_T(Q_{n,h}^{\mathfrak{g}}) = \frac{1}{2} + h,$$

the strength identity because the complier probability $p_n(x) = \mu_n w(x)/\kappa_n$ with $\mu_n = (t_0 \kappa_n/n)^{1/2}$ makes the transported first stage exactly $(t_0 \kappa_n/n)^{1/2}$, and the contrast identity because $Y(0) = 0$ while complier $Y(1)$ is Bernoulli with mean $1/2 + h$. The separation calculation of [definition 23](#) gives, for the same range of h ,

$$\|Q_{n,h}^{\mathfrak{g}} - Q_{n,0}^{\mathfrak{g}}\|_{\text{TV}} \leq 2\rho.$$

Step 2 (the two rowwise scalars). Let $(C_n) \in \mathfrak{C}^{\text{or}}(\mathfrak{g})$ be an arbitrary fixed-geometry honest oracle procedure, evaluated as in [definition 17](#) at the weight and propensity of $\mathfrak{g}$. For each row put

$$a_n = \inf_{P \in \mathcal{P}_n(\mathfrak{g})} P\{\theta_T(P) \in C_n\}, \quad r_n = \sup_{\substack{P \in \mathcal{P}_n(\mathfrak{g}) \\ t_n(P) \geq t_0}} \mathbb{E}_P[\lambda(C_n)],$$

and write, in the notation of [definition 7](#) restricted to the slice,

$$R_{\mathfrak{g}}((C_n), t_0) = \limsup_{n \rightarrow \infty} r_n = \limsup_{n \rightarrow \infty} \sup_{\substack{P \in \mathcal{P}_n(\mathfrak{g}) \\ t_n(P) \geq t_0}} \mathbb{E}_P[\lambda(C_n)],$$

so that $V_{\mathfrak{g}}^*(t_0) = \inf_{(C_n) \in \mathfrak{C}^{\text{or}}(\mathfrak{g})} R_{\mathfrak{g}}((C_n), t_0)$ by [definition 18](#). Each $P\{\theta_T(P) \in C_n\}$ is a probability, so $0 \leq a_n \leq 1$; each C_n is contained in Θ and $\lambda(\Theta) = 2$, so $0 \leq \mathbb{E}_P[\lambda(C_n)] \leq 2$ for every P and hence $0 \leq r_n \leq 2$.

Step 3 (the rowwise Le Cam inequality). Fix an index n at which Step 1 applies and put

$$I = \left[\frac{1}{2} - H(t_0), \frac{1}{2} + H(t_0) \right].$$

Because $H(t_0) \leq 1/4$ we have $I \subseteq [1/4, 3/4] \subseteq \Theta$, and $|u - 1/2| \leq H(t_0)$ for $u \in I$, so $Q_{n,u-1/2}^{\mathfrak{g}}$ is a member of the family of Step 1. Its target contrast is $\theta_T(Q_{n,u-1/2}^{\mathfrak{g}}) = u$, and it lies in $\mathcal{P}_n(\mathfrak{g})$, so the definition of a_n gives

$$a_n \leq Q_{n,u-1/2}^{\mathfrak{g}}\{u \in C_n\}, \quad u \in I.$$

Total variation compares any single event under the two laws, $|Q_{n,u-1/2}^{\mathfrak{g}}(A) - Q_{n,0}^{\mathfrak{g}}(A)| \leq \|Q_{n,u-1/2}^{\mathfrak{g}} - Q_{n,0}^{\mathfrak{g}}\|_{\text{TV}}$, so the bound of Step 1 transfers the display to the centered law:

$$a_n - 2\rho \leq Q_{n,0}^{\mathfrak{g}}\{u \in C_n\}, \quad u \in I.$$

The sample-by-parameter graph of C_n is measurable and $C_n \subseteq \Theta$, so Tonelli's theorem writes expected length as an inclusion integral, and integrating the previous display over I gives

$$\mathbb{E}_{Q_{n,0}^{\mathfrak{g}}}[\lambda(C_n)] = \int_{\Theta} Q_{n,0}^{\mathfrak{g}}\{u \in C_n\} du \geq \int_I Q_{n,0}^{\mathfrak{g}}\{u \in C_n\} du \geq 2H(t_0)(a_n - 2\rho),$$

the first inequality because the integrand is nonnegative and $I \subseteq \Theta$, the second because I has Lebesgue length $2H(t_0)$. Finally $Q_{n,0}^{\mathfrak{g}} \in \mathcal{P}_n(\mathfrak{g})$ and $t_n(Q_{n,0}^{\mathfrak{g}}) = t_0 \geq t_0$, so the centered law is admissible in the supremum defining r_n , and

$$2H(t_0)(a_n - 2\rho) \leq \mathbb{E}_{Q_{n,0}^{\mathfrak{g}}}[\lambda(C_n)] \leq r_n.$$

Step 4 (passage to the limit). The map $x \mapsto 2H(t_0)(x - 2\rho)$ is affine with nonnegative slope, hence monotone and continuous, and $a_n \in [0, 1]$, so it commutes with the $\liminf$; $r_n \in [0, 2]$, so $\liminf_n r_n \leq \limsup_n r_n$. Honesty in [definition 17](#) states $1 - \alpha \leq \liminf_n a_n$. Chaining these with the rowwise inequality of Step 3,

$$2H(t_0)(1 - \alpha - 2\rho) \leq \liminf_{n \rightarrow \infty} 2H(t_0)(a_n - 2\rho) \leq \liminf_{n \rightarrow \infty} r_n \leq \limsup_{n \rightarrow \infty} r_n = R_{\mathfrak{g}}((C_n), t_0).$$

Step 5 (calibrating the constant). Since $2\rho = (1 - \alpha)/4$,

$$1 - \alpha - 2\rho = \frac{3(1 - \alpha)}{4}.$$

Two cases, according to which term attains the minimum in $H(t_0)$. If $\rho t_0^{-1/2} \leq 1/4$, then $H(t_0) = \rho t_0^{-1/2} = (1 - \alpha)t_0^{-1/2}/8$ and

$$2H(t_0)(1 - \alpha - 2\rho) = 2 \cdot \frac{(1 - \alpha)t_0^{-1/2}}{8} \cdot \frac{3(1 - \alpha)}{4} = \frac{3(1 - \alpha)^2}{16} t_0^{-1/2} \geq c_{\alpha} \min\{1, t_0^{-1/2}\}.$$

If instead $1/4 \leq \rho t_0^{-1/2}$, then $H(t_0) = 1/4$ and

$$2H(t_0)(1 - \alpha - 2\rho) = \frac{1}{2} \cdot \frac{3(1 - \alpha)}{4} = \frac{3(1 - \alpha)}{8} \geq \frac{3(1 - \alpha)^2}{16} \geq c_{\alpha} \min\{1, t_0^{-1/2}\},$$

the middle inequality because $1 - \alpha \leq 1 \leq 2$. Hence in both cases

$$c_\alpha \min\{1, t_0^{-1/2}\} \leq R_{\mathbf{g}}((C_n), t_0) \quad \text{for every } (C_n) \in \mathfrak{C}^{\text{or}}(\mathbf{g}).$$

Step 6 (the honest class is nonempty, and the lower bound for the value). Let $D_n \equiv \Theta$ for every n . This is an oracle procedure: it is contained in Θ , its sample-by-parameter graph is the measurable set of pairs whose parameter coordinate lies in Θ , and it does not depend on the weight input at all. By [theorem 1](#), $\theta_T(P) \in \Theta$ for every $P \in \mathcal{P}_n(\mathbf{g})$, so

$$P\{\theta_T(P) \in D_n\} = 1 \quad \text{for every } P \in \mathcal{P}_n(\mathbf{g}),$$

whence the coverage row equals 1 at every index at which the slice is nonempty and $\liminf_n \inf_{P \in \mathcal{P}_n(\mathbf{g})} P\{\theta_T(P) \in D_n\} = 1 \geq 1 - \alpha$ by Step 0; thus $(D_n) \in \mathfrak{C}^{\text{or}}(\mathbf{g})$ and the class is nonempty. Taking the infimum of Step 5 over $\mathfrak{C}^{\text{or}}(\mathbf{g})$ in [definition 18](#),

$$c_\alpha \min\{1, t_0^{-1/2}\} \leq V_{\mathbf{g}}^\star(t_0).$$

Part B: the upper bound.

Step 7 (the oracle score-inversion rule). Write $L = L_\alpha = (8/(\alpha\varepsilon^2))^{1/2}$, let $e_{\mathbf{g}}$ be the index- n propensity of $\mathbf{g}$, and let

$$w_{\mathbf{g},n}(x) = \frac{dP_{T,\mathbf{g},n}}{dP_{S,\mathbf{g},n}^X}(x)$$

be the canonical density-ratio version determined by the two index- n covariate laws of $\mathbf{g}$; it is measurable, as is $e_{\mathbf{g}}$. With the inverse-propensity instrument score of [definition 1](#) formed at $e_{\mathbf{g}}$,

$$\begin{aligned} H_i &= \frac{Z_i}{e_{\mathbf{g}}(X_i)} - \frac{1 - Z_i}{1 - e_{\mathbf{g}}(X_i)}, & \hat{A}_n &= \frac{1}{n} \sum_{i=1}^n w_{\mathbf{g},n}(X_i) H_i Y_i, & \hat{B}_n &= \frac{1}{n} \sum_{i=1}^n w_{\mathbf{g},n}(X_i) H_i D_i, \\ \hat{\kappa}_n &= \frac{1}{n} \sum_{i=1}^n w_{\mathbf{g},n}(X_i)^2, \end{aligned}$$

define the procedure

$$C_n = \begin{cases} \emptyset, & n = 0, \\ \left\{ \vartheta \in \Theta : \left| \hat{A}_n - \vartheta \hat{B}_n \right| \leq L \sqrt{\frac{\hat{\kappa}_n}{n}} \right\}, & n \geq 1, \end{cases}$$

that is, the score-inversion set of [definition 14](#) evaluated at $(w_{\mathbf{g},n}, e_{\mathbf{g}})$. Each C_n is contained in Θ ; its sample-by-parameter graph is measurable because $\hat{A}_n$, $\hat{B}_n$, and $\hat{\kappa}_n$ are measurable functions of the source sample and $(s, \vartheta) \mapsto |\hat{A}_n - \vartheta \hat{B}_n| - L\sqrt{\hat{\kappa}_n/n}$ is jointly measurable; and it is unchanged if the supplied weight input is replaced by another version, since the rule does not read that input. Changing the single index $n = 0$ affects neither the liminf in [definition 17](#) nor the limsup in [definition 18](#).

Step 8 (what the slice fixes at a row). Fix $n \geq 1$ and $P \in \mathcal{P}_n(\mathbf{g})$. By [definition 16](#) the source covariate marginal, the target covariate law, and the propensity of P are exactly the index- n components of $\mathbf{g}$, so the two density ratios and the two propensities coincide as functions:

$$w_{\mathbf{g},n} = w = \frac{dP_T}{dP_S^X}, \quad e_{\mathbf{g}} = e.$$

Thus $\hat{A}_n, \hat{B}_n, \hat{\kappa}_n$ are formed with P 's own oracle inputs. By [theorem 1](#),

$$\theta_T \in \Theta, \quad \mu_n = P_n^F \{D(1) = 1, D(0) = 0 \mid S = 0\}, \quad \frac{\mu_{Y,n}}{\mu_n} = \theta_T,$$

and target complier positivity, [assumption 11](#), therefore gives $\mu_n > 0$. Transport domination, [assumption 12](#), makes w a genuine density ratio, so change of measure gives

$$\mathbb{E}_{P_S^X}[w(X)] = P_T(\mathcal{X}) = 1,$$

and Cauchy–Schwarz then yields

$$\kappa_n = \mathbb{E}_{P_S^X}[w(X)^2] \geq \left(\mathbb{E}_{P_S^X}[w(X)]\right)^2 = 1.$$

Step 9 (the pullout identity). Let F be a function of the source observation and d a measurable function of the covariate, both integrable under P_S and P_S^X respectively, such that $w(X)F$ is P_S -integrable, $w d$ is P_S^X -integrable, and

$$\int_{\{O: X \in A\}} F(O) dP_S(O) = \int_A d(x) dP_S^X(x) \quad \text{for every measurable } A \subseteq \mathcal{X}.$$

Then

$$\int w(X)F(O) dP_S(O) = \int w(x)d(x) dP_S^X(x).$$

Indeed the hypothesis says exactly that $d(X)$ is a version of $\mathbb{E}_{P_S}[F \mid X]$; since $w(X)$ is a measurable function of X alone and both $w(X)F(O)$ and $w d$ are integrable under the two stated laws, $w(X)$ may be pulled out of that conditional expectation, and the tower property gives the display.

Step 10 (first-stage moments and the bad-slope event). The representation identities of [theorem 1](#) are the hypothesis of Step 9 with $F = H(O)D$, $d = \Delta_D$, and with $F = H(O)Y$, $d = \Delta_Y$. Hence

$$\int w(X)H(O)D dP_S = \mathbb{E}_{P_S^X}[w(X)\Delta_D(X)] = \mu_n, \quad \int w(X)H(O)Y dP_S = \mathbb{E}_{P_S^X}[w(X)\Delta_Y(X)] = \mu_{Y,n}.$$

Instrument overlap, [assumption 4](#), gives $|H(O)| \leq 1/\varepsilon$ P_S -almost surely, and $D \in \{0, 1\}$, so

$$\text{Var}_{P_S}(w(X)H(O)D) \leq \int w(X)^2 H(O)^2 D^2 dP_S \leq \frac{1}{\varepsilon^2} \int w(X)^2 dP_S = \frac{\kappa_n}{\varepsilon^2}.$$

Since $\hat{B}_n$ is the average of n independent copies of $w(X)H(O)D$,

$$\mathbb{E}_P[\hat{B}_n] = \mu_n, \quad \text{Var}_P(\hat{B}_n) \leq \frac{\kappa_n}{\varepsilon^2 n},$$

and Chebyshev's inequality at the deviation $\mu_n/2$ gives

$$P\left(\frac{\mu_n}{2} < \left|\hat{B}_n - \mu_n\right|\right) \leq \frac{\kappa_n/(\varepsilon^2 n)}{(\mu_n/2)^2} = \frac{4}{\varepsilon^2 t_n}.$$

Step 11 (pointwise expected length). Write $r = L\sqrt{\widehat{\kappa}_n/n}$, so that $C_n = \{\vartheta \in \Theta : |\widehat{A}_n - \vartheta \widehat{B}_n| \leq r\}$. When $\widehat{B}_n \neq 0$ this set is the intersection of Θ with the interval of centre $\widehat{A}_n/\widehat{B}_n$ and length $2r/|\widehat{B}_n|$; and $\lambda(C_n) \leq \lambda(\Theta) = 2$ always. Hence

$$\lambda(C_n) \leq \min \left\{ 2, \frac{2r}{|\widehat{B}_n|} \right\}.$$

On the event $\{|\widehat{B}_n - \mu_n| \leq \mu_n/2\}$ we have $|\widehat{B}_n| \geq \mu_n/2$ and therefore $\lambda(C_n) \leq 4L\sqrt{\widehat{\kappa}_n/n}/\mu_n$; on its complement we use $\lambda(C_n) \leq 2$. Since $\widehat{\kappa}_n \geq 0$ with $\mathbb{E}_P[\widehat{\kappa}_n] = \kappa_n$, concavity of the square root gives $\mathbb{E}_P[\sqrt{\widehat{\kappa}_n}] \leq \sqrt{\kappa_n}$, and combining with the tail bound of Step 10,

$$\mathbb{E}_P[\lambda(C_n)] \leq \frac{4L}{\mu_n} \sqrt{\frac{\kappa_n}{n}} + 2 \cdot \frac{4}{\varepsilon^2 t_n} = 4L t_n^{-1/2} + \frac{8}{\varepsilon^2 t_n},$$

using $\mu_n^{-1} \sqrt{\kappa_n/n} = t_n^{-1/2}$. If $t_n \leq 1$ then $\min\{1, t_n^{-1/2}\} = 1$ and $\mathbb{E}_P[\lambda(C_n)] \leq 2$ already suffices; if $t_n > 1$ then $t_n^{-1} \leq t_n^{-1/2} = \min\{1, t_n^{-1/2}\}$. In both cases

$$\mathbb{E}_P[\lambda(C_n)] \leq \max \left\{ 2, 4L + \frac{8}{\varepsilon^2} \right\} \min\{1, t_n^{-1/2}\}.$$

At $n = 0$ the reported set is empty, so its expected length vanishes; moreover the restriction $t_n \geq t_0 > 0$ forces $n \geq 1$, so the index $n = 0$ never enters the strength-restricted supremum.

Step 12 (the coverage deficit sequence). Define

$$\delta_n = \begin{cases} 1, & n = 0, \\ 16k_n^2/n, & n > 0. \end{cases}$$

For $n > 0$, $\delta_n = 16(k_n/\sqrt{n})^2$, so $k_n/\sqrt{n} \rightarrow 0$ gives $\delta_n \rightarrow 0$.

Step 13 (rowwise coverage). Fix $n \geq 1$ and $P \in \mathcal{P}_n(\mathfrak{g})$, and set

$$\psi(O) = w(X)H(O) \{Y - \theta_T D\}, \quad S_n = \widehat{A}_n - \theta_T \widehat{B}_n = \frac{1}{n} \sum_{i=1}^n \psi(O_i^S).$$

By the two moment identities of Step 10 and $\mu_{Y,n}/\mu_n = \theta_T$ from Step 8,

$$\int \psi dP_S = \mu_{Y,n} - \theta_T \mu_n = 0.$$

Overlap gives $|H(O)| \leq 1/\varepsilon$, while $Y \in [0, 1]$, $D \in \{0, 1\}$, and $\theta_T \in \Theta$ give $|Y - \theta_T D| \leq 2$; hence $\psi^2 \leq 4w(X)^2/\varepsilon^2$ pointwise, so

$$\text{Var}_{P_S}(\psi) \leq \int \psi^2 dP_S \leq \frac{4\kappa_n}{\varepsilon^2}, \quad \text{Var}_P(S_n) \leq \frac{4\kappa_n}{\varepsilon^2 n}, \quad \mathbb{E}_P[S_n] = 0.$$

For the dispersion statistic, the weight envelope $0 \leq w \leq 2k_n$ of [assumption 13](#) gives the pointwise fourth-moment bound $w^4 \leq 4k_n^2 w^2$, so

$$\mathbb{E}_P[\widehat{\kappa}_n] = \kappa_n, \quad \text{Var}_P(\widehat{\kappa}_n) \leq \frac{4k_n^2 \kappa_n}{n},$$

and Chebyshev's inequality at the deviation $\kappa_n/2$ yields

$$P\left(\widehat{\kappa}_n < \frac{\kappa_n}{2}\right) \leq \frac{4k_n^2\kappa_n/n}{(\kappa_n/2)^2} = \frac{16k_n^2}{n\kappa_n} \leq \frac{16k_n^2}{n} = \delta_n,$$

the last step by $\kappa_n \geq 1$ from Step 8. Next put

$$a = L\sqrt{\frac{\kappa_n}{2n}}, \quad \text{so that} \quad a^2 = \frac{L^2\kappa_n}{2n} = \frac{4\kappa_n}{\alpha\varepsilon^2 n} \quad \text{since } L^2 = \frac{8}{\alpha\varepsilon^2},$$

and Chebyshev's inequality applied to the mean-zero statistic S_n gives

$$P\left(a < \left|\widehat{A}_n - \theta_T \widehat{B}_n\right|\right) \leq \frac{4\kappa_n/(\varepsilon^2 n)}{a^2} = \alpha.$$

On the complement of the two bad events $\{\widehat{\kappa}_n < \kappa_n/2\}$ and $\{a < |S_n|\}$ we have $\widehat{\kappa}_n \geq \kappa_n/2$, hence $\sqrt{\kappa_n/(2n)} \leq \sqrt{\widehat{\kappa}_n/n}$, and therefore

$$\left|\widehat{A}_n - \theta_T \widehat{B}_n\right| \leq a = L\sqrt{\frac{\kappa_n}{2n}} \leq L\sqrt{\frac{\widehat{\kappa}_n}{n}},$$

so $\theta_T \in C_n$ because also $\theta_T \in \Theta$. Both bad events are events of the source sample, and the two-sample law is the product of the source and target sampling laws, so the union bound gives, for every $P \in \mathcal{P}_n(\mathfrak{g})$,

$$1 - \alpha - \delta_n \leq P\{\theta_T(P) \in C_n\}.$$

At $n = 0$ the left-hand side is $-\alpha < 0$, so the display holds there as well.

Step 14 (honesty). Coverage values lie in $[0, 1]$, the slice is nonempty for all sufficiently large n by Step 0, and $\delta_n \rightarrow 0$ by Step 12. Taking the rowwise infimum of Step 13 and then the liminf,

$$1 - \alpha \leq \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{P}_n(\mathfrak{g})} P\{\theta_T(P) \in C_n\},$$

which is exactly the requirement of [definition 17](#), so $(C_n) \in \mathfrak{C}^{\text{or}}(\mathfrak{g})$.

Step 15 (frontier risk and the value). Recall the upper-bound constant of the statement,

$$C_\alpha = \max\left\{2, 4L_\alpha + \frac{8}{\varepsilon^2}\right\}.$$

The map $t \mapsto \min\{1, t^{-1/2}\}$ is nonincreasing on $(0, \infty)$, so $t_n \geq t_0$ gives $\min\{1, t_n^{-1/2}\} \leq \min\{1, t_0^{-1/2}\}$; combined with Step 11, for every n and every $P \in \mathcal{P}_n(\mathfrak{g})$ with $t_n(P) \geq t_0$,

$$\mathbb{E}_P[\lambda(C_n)] \leq C_\alpha \min\{1, t_0^{-1/2}\}.$$

Taking the supremum over such P and then the limsup in n ,

$$R_{\mathfrak{g}}((C_n), t_0) \leq C_\alpha \min\{1, t_0^{-1/2}\}.$$

Expected lengths are nonnegative, so $R_{\mathfrak{g}}((D_n), t_0) \geq 0$ for every $(D_n) \in \mathfrak{C}^{\text{or}}(\mathfrak{g})$ and the infimum in [definition 18](#) is over a set bounded below; since (C_n) is admissible for that infimum by Step 14,

$$V_{\mathfrak{g}}^*(t_0) \leq R_{\mathfrak{g}}((C_n), t_0) \leq C_\alpha \min\{1, t_0^{-1/2}\}.$$

Combining Steps 6 and 15 gives

$$c_\alpha \min\{1, t_0^{-1/2}\} \leq V_{\mathfrak{g}}^*(t_0) \leq C_\alpha \min\{1, t_0^{-1/2}\},$$

and $t_0 > 0$ was arbitrary. The constants $c_\alpha = 3(1 - \alpha)^2/16$ and $C_\alpha = \max\{2, 4L_\alpha + 8\varepsilon^{-2}\}$ were built only from α and ε , never from $\mathfrak{g}$, so they are uniform over all admissible deterministic geometries. $\square$

Proof of [theorem 5](#). Fix the admissible geometry array $\mathfrak{g} = (P_{S,n}^X, w_n, e_n)_{n \geq 1}$. We prove the three asserted conclusions in the order in which they appear.

1. Fix n . By [definition 15](#) the geometry second moment at index n is

$$\kappa_n = \int w_n(x)^2 dP_{S,n}^X(x),$$

and $P_{S,n}^X$ is a probability law on $\mathcal{X}$. The unit-weight hypothesis states $w_n(x) = 1$ for $P_{S,n}^X$ -almost every x , hence also

$$w_n(x)^2 = 1 \quad P_{S,n}^X\text{-a.s.},$$

and two functions that agree almost everywhere have the same integral, so

$$\kappa_n = \int w_n(x)^2 dP_{S,n}^X(x) = \int 1 dP_{S,n}^X(x) = 1.$$

For the equality of covariate laws, let $A \subseteq \mathcal{X}$ be measurable. The change-of-measure clause of [definition 15](#) states

$$P_{T,n}(A) = \int_A w_n(x) dP_{S,n}^X(x).$$

A $P_{S,n}^X$ -null set is also null for the restriction of $P_{S,n}^X$ to A , so the almost-sure identity $w_n = 1$ persists after that restriction, and therefore

$$P_{T,n}(A) = \int_A 1 dP_{S,n}^X(x) = P_{S,n}^X(A).$$

Two measures that assign the same value to every measurable set are equal, so

$$P_{T,n} = P_{S,n}^X.$$

2. Let $P \in \mathcal{P}_n(\mathfrak{g})$. Write

$$w_n^P = \frac{dP_{T,n}}{dP_{S,n}^X}$$

for the nonnegative Radon–Nikodym transport weight induced by the laws of P itself, and let

$$\kappa_n^P = \int \{w_n^P(x)\}^2 dP_{S,n}^X(x), \quad \mu_n = \int w_n^P(x) \Delta_D(x) dP_{S,n}^X(x)$$

be the associated dispersion and transported first stage of [definition 2](#), so that the effective strength of P at index n is

$$t_n = \frac{n\mu_n^2}{\kappa_n^P}.$$

By [definition 16](#), membership $P \in \mathcal{P}_n(\mathfrak{g})$ means that $P \in \mathcal{P}_n$ and that the index- n triple of P is the index- n component of $\mathfrak{g}$; in particular the source covariate marginal of P is $P_{S,n}^X$ and

$$w_n^P(x) = w_n(x) \quad P_{S,n}^X\text{-a.s.}$$

(the almost-sure form is the correct one, since a density ratio is pinned only up to $P_{S,n}^X$ -null sets). Integrating the squares of two functions that agree $P_{S,n}^X$ -almost everywhere against the common law $P_{S,n}^X$ gives

$$\kappa_n^P = \int \{w_n^P(x)\}^2 dP_{S,n}^X(x) = \int w_n(x)^2 dP_{S,n}^X(x) = \kappa_n.$$

By the first step $\kappa_n = 1$, so $\kappa_n^P = 1$, and substituting this into the display for t_n yields

$$t_n = \frac{n\mu_n^2}{\kappa_n^P} = \frac{n\mu_n^2}{1} = n\mu_n^2.$$

3. Fix $t_0 > 0$. We verify the hypotheses of [theorem 4](#) for the same $\mathfrak{g}$, c , ε , α , k_n , and N_n . Its target-sample-ratio hypothesis is $c > 0$ with $N_n/n \rightarrow c$; its envelope-scale hypothesis is $k_n > 0$ for every n , $k_n \rightarrow \infty$, and $k_n/\sqrt{n} \rightarrow 0$; its overlap and noncoverage hypotheses are $0 < \varepsilon < 1/2$ and $0 < \alpha < 1$; and its geometry hypothesis is admissibility of $\mathfrak{g}$ at scale k_n and overlap constant ε . These are exactly the standing hypotheses of the present statement.

Its remaining hypothesis is the fixed-geometry slice condition: for every n and every transported array whose index- n law lies in $\mathcal{P}_n(\mathfrak{g})$, the conditions of [assumptions 3, 4 and 13 to 15](#) hold. This holds because [definition 16](#) defines $\mathcal{P}_n(\mathfrak{g})$ as a subset of $\mathcal{P}_n$, and each of those five conditions is one of the requirements listed in [definition 3](#); membership in $\mathcal{P}_n$ therefore supplies all five directly. (The unit-weight hypothesis is not needed here: this part of the conclusion is the fixed-geometry frontier for $\mathfrak{g}$ verbatim.)

Applying [theorem 4](#) at this t_0 therefore gives, with

$$L_\alpha = \sqrt{\frac{8}{\alpha\varepsilon^2}}, \quad c_\alpha = \frac{3(1-\alpha)^2}{16}, \quad C_\alpha = \max\left\{2, 4L_\alpha + \frac{8}{\varepsilon^2}\right\},$$

the two-sided bound on the fixed-geometry minimax value of [definition 18](#),

$$c_\alpha \min\{1, t_0^{-1/2}\} \leq V_\mathfrak{g}^\star(t_0) \leq C_\alpha \min\{1, t_0^{-1/2}\},$$

which is the asserted conclusion since $t_0 > 0$ was arbitrary.

□

Proof of [theorem 2](#). 1. *The constants.* Set

$$c_0 = \frac{3(1-\alpha)^2}{16}, \quad t_c = 1.$$

Since $0 < \alpha < 1$ we have $1 - \alpha > 0$, hence $c_0 > 0$; and $t_c = 1 > 0$. Both numbers are fixed at this point, before any covariate space, any arrays N_n, k_n , any transported law, and any threshold are named. Fix from here on a measurable covariate space $\mathcal{X}$, arrays N_n, k_n with $N_n/n \rightarrow c$, $k_n > 0$ for every n , $k_n \rightarrow \infty$, and $k_n/\sqrt{n} \rightarrow 0$, and a threshold $t_0 > 0$.

2. *A uniform finite-cell geometry.* By the measurable-carrier hypothesis, choose for each n distinct points $x_{n,1}, \dots, x_{n,k_n} \in \mathcal{X}$ whose singletons $\{x_{n,i}\}$ are measurable, and set

$$\nu_n = \frac{1}{k_n} \sum_{i=1}^{k_n} \delta_{x_{n,i}}.$$

Since $k_n > 0$, $\nu_n(\mathcal{X}) = k_n \cdot k_n^{-1} = 1$, so ν_n is a probability law; because the carriers are distinct,

$$\nu_n(\{x_{n,i}\}) = \frac{1}{k_n} \quad (1 \leq i \leq k_n), \quad \nu_n(\{x_{n,1}, \dots, x_{n,k_n}\}) = 1.$$

Define the deterministic array $\mathbf{g} = (P_S^X, w, e)_{n \geq 1}$ by

$$P_{S,n}^X = \nu_n, \quad P_{T,n} = \nu_n, \quad w_n(x) = 1, \quad e_n(x) = \frac{1}{2} \quad (x \in \mathcal{X}).$$

3. $\mathbf{g} \in \mathcal{G}$. We verify each clause of [definition 15](#). Source and target laws are the probability law ν_n . Overlap holds because $0 < \varepsilon < 1/2$:

$$\varepsilon \leq \frac{1}{2} = e_n(x) \leq 1 - \varepsilon.$$

The envelope holds because k_n is a positive integer, so $k_n \geq 1$ and

$$0 \leq w_n(x) = 1 \leq 2k_n.$$

Normalization and the second-moment bound follow from $\nu_n(\mathcal{X}) = 1$:

$$\mathbb{E}_{P_{S,n}^X} [w_n(X)] = \int 1 d\nu_n = 1, \quad \kappa_n = \mathbb{E}_{P_{S,n}^X} [w_n(X)^2] = \int 1 d\nu_n = 1 \leq k_n.$$

Finally, the change-of-measure clause reduces to integrating the constant weight: for every measurable $A \subseteq \mathcal{X}$,

$$\mathbb{E}_{P_{S,n}^X} [w_n(X) \mathbf{1}\{X \in A\}] = \int_A 1 d\nu_n = \nu_n(A) = P_{T,n}(A).$$

4. *The fixed-geometry slice lies in the finite-cell submodel:* for every n ,

$$\mathcal{P}_n(\mathbf{g}) \subseteq \mathcal{N}_n.$$

Indeed, let $P \in \mathcal{P}_n(\mathbf{g})$. By [definition 16](#) we have $P \in \mathcal{P}_n$, so by [definition 22](#) the only condition left to check is [assumption 16](#), and it holds with the carriers of the previous step: the source covariate marginal of P is $P_{S,n}^X = \nu_n$, a probability law carried by the k_n distinct measurable singletons $\{x_{n,i}\}$ with

$$P_{S,n}^X(\{x_{n,i}\}) = \frac{1}{k_n}, \quad P_{S,n}^X(\{x_{n,1}, \dots, x_{n,k_n}\}) = 1,$$

and the propensity of P is the geometry propensity, so that

$$e(X) = \frac{1}{2} \quad P_S\text{-almost surely.}$$

5. *The oracle honest class is nonempty.* Let D^{full} be the procedure returning the whole parameter space at every input,

$$D_n^{\text{full}}\left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w, e\right) = \Theta,$$

which satisfies the structural requirements in [definition 6](#): its values are subsets of Θ ; for each fixed measurable (w, e) its sample-by-parameter graph is the preimage of the closed interval $\Theta = [-1, 1]$ under $\left(((O_i^S)_i, (X_j^T)_j), \vartheta\right) \mapsto \vartheta$, hence measurable; and its output does not depend on w , so two almost surely equal weight versions give the same set. For every n and every $P \in \mathcal{P}_n$, [theorem 1](#) gives $\theta_T \in \Theta$, so the covering event has full probability:

$$P\left(\theta_T \in D_n^{\text{full}}\right) = 1.$$

Moreover $\mathcal{P}_n \neq \emptyset$ for every n , because the measurable carriers $x_{n,1}, \dots, x_{n,k_n}$ together with the size and cell-scale hypotheses carry a uniform finite-cell transported law, which lies in $\mathcal{N}_n \subseteq \mathcal{P}_n$; hence the first branch of the coverage functional in [definition 6](#) applies and

$$\inf_{P \in \mathcal{P}_n} P\left(\theta_T \in D_n^{\text{full}}\right) = 1 \quad \text{for every } n, \quad \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{P}_n} P\left(\theta_T \in D_n^{\text{full}}\right) = 1 \geq 1 - \alpha,$$

the last inequality because $\alpha > 0$. Thus $D^{\text{full}} \in \mathfrak{C}^{\text{or}}$, and the infima defining $V^*(t_0)$ and $V_{\mathcal{N}}^*(t_0)$ in [definitions 8](#) and [9](#) range over a nonempty class.

6. *The fixed-geometry lower bound.* Under $\mathfrak{g}$ the transport weight is identically one, so in particular $w_n(X) = 1$ holds $P_{S,n}^X$ -almost surely for every n , and [theorem 5](#) applies to $\mathfrak{g}$. Its lower-bound constant is $c_\alpha = 3(1 - \alpha)^2/16 = c_0$, so for every $t_0 > 0$,

$$c_0 \min\{1, t_0^{-1/2}\} \leq V_{\mathfrak{g}}^*(t_0),$$

with $V_{\mathfrak{g}}^*(t_0)$ the fixed-geometry value of [definition 18](#).

7. *First conclusion.* Let $(C_n) \in \mathfrak{C}^{\text{or}}$ be arbitrary. Evaluated at the fixed oracle inputs (w_n, e_n) of $\mathfrak{g}$, the sequence (C_n) is itself a member of $\mathfrak{C}^{\text{or}}(\mathfrak{g})$, and its fixed-geometry risk does not exceed its global risk:

$$V_{\mathfrak{g}}^*(t_0) \leq \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}_n(\mathfrak{g}): t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)] \leq R((C_n), t_0).$$

The first inequality is the definition of $V_{\mathfrak{g}}^*(t_0)$ once slice honesty is known, and slice honesty holds because on each row $P \in \mathcal{P}_n(\mathfrak{g})$ the geometry's declared weight w_n is a version of $dP_{T,n}/dP_{S,n}^X$ under P , so by the weight-invariance requirement of [definition 6](#) the fixed-geometry evaluation of C_n coincides almost surely with its evaluation at P 's canonical density ratio, whence the two coverage probabilities and the two expected lengths agree row by row. The second inequality holds because $\mathcal{P}_n(\mathfrak{g}) \subseteq \mathcal{P}_n$, so the strength-restricted supremum in [definition 7](#) is over a larger family at every n . Combining with the previous step and taking the infimum over $(C_n) \in \mathfrak{C}^{\text{or}}$ in [definition 8](#),

$$c_0 \min\{1, t_0^{-1/2}\} \leq V^*(t_0).$$

8. *Two length bounds.* For every $A \subseteq \mathbb{R}$ the length criterion of [definition 7](#) is Lebesgue measure after intersecting with $\Theta = [-1, 1]$, so

$$0 \leq \lambda(A) \leq \lambda(\Theta) = 2.$$

Consequently, for any oracle procedure (E_n) and any n , each expected length $\mathbb{E}_P[\lambda(E_n)]$ lies in $[0, 2]$, and therefore, with the convention that a supremum over an empty family is zero,

$$0 \leq \sup_{P \in \mathcal{P}_n(\mathfrak{g}): t_n \geq t_0} \mathbb{E}_P[\lambda(E_n)], \quad \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(E_n)] \leq 2.$$

Taking limits superior, the fixed-geometry risk of every oracle procedure is nonnegative, so the family of fixed-geometry risks over $\mathfrak{C}^{\text{or}}(\mathfrak{g})$ is bounded below and $V_{\mathfrak{g}}^*(t_0)$ is the infimum of that family; and the finite-cell risk rows are bounded above by 2. These two facts license the comparison of limits superior used below.

9. *A slice procedure attached to a finite-cell oracle.* Fix $(C_n) \in \mathfrak{C}^{\text{or}}$. Write

$$w_n^{\mathfrak{g}} = \frac{dP_{T,n}}{dP_{S,n}^X}$$

for the canonical nonnegative density-ratio version of the geometry $\mathfrak{g}$, which is measurable, and define the oracle procedure (D_n) that discards its own weight and propensity inputs and feeds (C_n) the geometry's canonical inputs:

$$D_n\left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w, e\right) = C_n\left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w_n^{\mathfrak{g}}, e_n\right).$$

Its values lie in Θ , its graph is measurable because $w_n^{\mathfrak{g}}$ and e_n are measurable, and it is trivially invariant to the supplied weight. For every n , every $P \in \mathcal{P}_n(\mathfrak{g})$, and every realized sample, the fixed-geometry evaluation of D_n and the evaluation of C_n at P 's own oracle inputs are the same set,

$$D_n\left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w_n, e_n\right) = C_n\left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, \frac{dP_T}{dP_S^X}, e\right),$$

because on the slice the source law, the target law, and the propensity of P are exactly $P_{S,n}^X$, $P_{T,n}$, and e_n , so P 's canonical density ratio is $w_n^{\mathfrak{g}}$. Hence coverage probabilities and expected lengths agree row by row on the slice:

$$P(\theta_T \in D_n) = P(\theta_T \in C_n), \quad \mathbb{E}_P[\lambda(D_n)] = \mathbb{E}_P[\lambda(C_n)], \quad P \in \mathcal{P}_n(\mathfrak{g}).$$

10. $(D_n) \in \mathfrak{C}^{\text{or}}(\mathfrak{g})$. The least-favorable construction of [definition 19](#) at the threshold t_0 , applied to the admissible geometry $\mathfrak{g}$, supplies for all sufficiently large n a law

$$P \in \mathcal{P}_n(\mathfrak{g}) \quad \text{with} \quad t_n \geq t_0,$$

so in particular the slice $\mathcal{P}_n(\mathfrak{g})$ is eventually nonempty. For such n and every $P \in \mathcal{P}_n(\mathfrak{g}) \subseteq \mathcal{P}_n$, the row identity of the previous step gives $\inf_{Q \in \mathcal{P}_n} Q(\theta_T \in C_n) \leq P(\theta_T \in C_n) = P(\theta_T \in D_n)$; taking the infimum over the slice,

$$\inf_{Q \in \mathcal{P}_n} Q(\theta_T \in C_n) \leq \inf_{P \in \mathcal{P}_n(\mathfrak{g})} P(\theta_T \in D_n) \quad \text{for all large } n.$$

Coverage probabilities lie in $[0, 1]$, so both sides are bounded sequences and the inequality passes to the limit inferior; combined with the honesty of (C_n) ,

$$1 - \alpha \leq \liminf_{n \rightarrow \infty} \inf_{Q \in \mathcal{P}_n} Q(\theta_T \in C_n) \leq \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{P}_n(\mathfrak{g})} P(\theta_T \in D_n),$$

which is the honesty requirement of [definition 17](#).

11. *Second conclusion.* For all sufficiently large n the slice $\mathcal{P}_n(\mathfrak{g})$ contains a law of strength at least t_0 , and by step 4 every such law lies in $\mathcal{N}_n$; together with the equality of expected lengths on the slice this gives, for all large n ,

$$\sup_{P \in \mathcal{P}_n(\mathfrak{g}): t_n \geq t_0} \mathbb{E}_P[\lambda(D_n)] \leq \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)].$$

By the two length bounds of step 8 the left side is nonnegative and the right side is at most 2, so the comparison passes to the limits superior:

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}_n(\mathfrak{g}): t_n \geq t_0} \mathbb{E}_P[\lambda(D_n)] \leq \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)].$$

Since $(D_n) \in \mathfrak{C}^{\text{or}}(\mathfrak{g})$, the left side dominates the infimum defining $V_{\mathfrak{g}}^*(t_0)$ in [definition 18](#), so with step 6,

$$c_0 \min\{1, t_0^{-1/2}\} \leq V_{\mathfrak{g}}^*(t_0) \leq \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)].$$

As $(C_n) \in \mathfrak{C}^{\text{or}}$ was arbitrary, taking the infimum over $\mathfrak{C}^{\text{or}}$ in [definition 9](#) yields

$$c_0 \min\{1, t_0^{-1/2}\} \leq V_{\mathcal{N}}^*(t_0).$$

12. *Third conclusion and uniformity.* Let $0 < t_0 \leq t_c = 1$. Since the exponent $-1/2$ is nonpositive and $0 < t_0 \leq 1$, we get $t_0^{-1/2} \geq 1$ and therefore

$$\min\{1, t_0^{-1/2}\} = 1,$$

so the bound of step 7 reads $c_0 \leq V^*(t_0)$. Finally, c_0 and t_c were fixed in step 1, before the covariate space, the arrays N_n, k_n , the transported laws, and the threshold t_0 were chosen, and no step above constrains a row of the model beyond its membership in $\mathcal{P}_n$ (or $\mathcal{N}_n$) and the strength condition $t_n = n\mu_n^2/\kappa_n \geq t_0$. The same two constants therefore serve uniformly over all decompositions of the effective strength t_0 into a transported first-stage mean μ_n and a weight dispersion κ_n . □

Proof of [theorem 3](#). 1. **Construction.** Because $\alpha \in (0, 1)$ and $\varepsilon \in (0, 1/2)$, the constant

$$L_\alpha = \left(\frac{8}{\alpha \varepsilon^2} \right)^{1/2}$$

is a positive real number, and

$$C_0 = \max \left\{ 2, 4L_\alpha + \frac{8}{\varepsilon^2} \right\} \geq 2 > 0.$$

For each n , each source sample $(O_i^S)_{i=1}^n$, each target covariate sample $(X_j^T)_{j=1}^{N_n}$, each measurable nonnegative $w : \mathcal{X} \rightarrow \mathbb{R}$ and each measurable $e : \mathcal{X} \rightarrow \mathbb{R}$, let C_n be the score-inversion set of [definition 14](#) formed from those inputs,

$$C_n = \left\{ \vartheta \in \Theta : \left| \hat{A}_n - \vartheta \hat{B}_n \right| \leq L_\alpha \sqrt{\hat{\kappa}_n/n} \right\}, \quad \hat{A}_n = \frac{1}{n} \sum_{i=1}^n w(X_i) H_i Y_i, \quad \hat{B}_n = \frac{1}{n} \sum_{i=1}^n w(X_i) H_i D_i,$$

$$\hat{\kappa}_n = \frac{1}{n} \sum_{i=1}^n w(X_i)^2, \quad H_i = \frac{Z_i}{e(X_i)} - \frac{1 - Z_i}{1 - e(X_i)}.$$

Only the source sample enters; the target covariate sample is not used. We check the three requirements that [definition 6](#) imposes on an oracle procedure.

Range. $C_n \subseteq \Theta$, because membership is defined by intersecting the score restriction with Θ .

Measurable graph. For measurable w and e the multiplier H is measurable, hence so are the sample functions $\hat{A}_n, \hat{B}_n, \hat{\kappa}_n$; the sample-by-parameter graph is

$$\left\{ \left(\left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n} \right), \vartheta \right) : \vartheta \in \Theta \right\} \cap \left\{ \left(\left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n} \right), \vartheta \right) : \left| \hat{A}_n - \vartheta \hat{B}_n \right| \leq L_\alpha \sqrt{\hat{\kappa}_n/n} \right\},$$

an intersection of a preimage of the closed interval Θ with a sublevel set of a jointly measurable function of the sample and ϑ .

Weight-version invariance. The three statistics $\hat{A}_n, \hat{B}_n, \hat{\kappa}_n$ depend on w only through the realized vector $(w(X_1), \dots, w(X_n))$, so

$$w(X_i) = w'(X_i) \text{ for all } i \leq n \implies C_n \left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w, e \right) = C_n \left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w', e \right).$$

If $w = w'$ holds P_S^X -almost surely, then, each X_i having marginal law P_S^X and the source sample being an independent product, the antecedent holds almost surely under the two-sample law; hence the two evaluations agree almost surely.

Finally we record that the comparison class is eventually nonempty. Let $\mathbf{g} \in \mathcal{G}$ be the admissible geometry supplied by the hypothesis. For every $t_0 > 0$, the least-favorable family of [definition 19](#) at geometry $\mathbf{g}$ and threshold t_0 supplies, for all sufficiently large n , a law

$$P \in \mathcal{P}_n(\mathbf{g}) \quad \text{with} \quad t_n \geq t_0;$$

since $\mathcal{P}_n(\mathbf{g}) \subseteq \mathcal{P}_n$, the full class $\mathcal{P}_n$ is nonempty at all sufficiently large n .

2. **Weighted conditional-expectation reduction.** The proof uses the following pull-out identity. Suppose F is a source-observation function, d is a covariate function, and w is a measurable transport weight, with $F, d, w(X)F$ and wd integrable under the indicated laws. If, for every measurable $A \subseteq \mathcal{X}$,

$$\int_{\{O: X \in A\}} F(O) dP_S(O) = \int_A d(x) dP_S^X(x),$$

then

$$\int w(X) F(O) dP_S(O) = \int w(x) d(x) dP_S^X(x).$$

Indeed, the set-integral identity identifies $d(X)$ with the conditional expectation of F given X ; multiplying by the X -measurable function $w(X)$ and integrating gives the displayed equality. The measurability and integrability requirements are supplied by overlap, the weight envelope, and the bounded source-observation variables.

3. **Pointwise expected length.** Fix $n \geq 1$ and $P \in \mathcal{P}_n$, and write $\mu_{Y,n}, \mu_n, \kappa_n, t_n$ for the functionals of [definition 2](#) at P and $\theta_n = \theta_T$ for the target complier effect of [definition 5](#) at P . By [theorem 1](#),

$$\mu_n = P_n^F\{D(1) = 1, D(0) = 0 \mid S = 0\}, \quad \frac{\mu_{Y,n}}{\mu_n} = \theta_n, \quad \theta_n \in \Theta,$$

and target complier positivity makes the first display positive, so $\mu_n > 0$. Under transport domination $w = dP_T/dP_S^X$ satisfies $\mathbb{E}_{P_S^X}[w(X)] = P_T(\mathcal{X}) = 1$, whence by the Cauchy–Schwarz inequality

$$1 = \left(\mathbb{E}_{P_S^X}[w(X)]\right)^2 \leq \mathbb{E}_{P_S^X}[w(X)^2] = \kappa_n,$$

so $\kappa_n \geq 1 > 0$.

Consider the per-observation weighted receipt score

$$\varphi(O) = w(X)H(O)D.$$

Instrument overlap $\varepsilon \leq e(X) \leq 1 - \varepsilon$ gives $|H(O)| \leq 1/\varepsilon$; together with $D \in \{0, 1\}$ and the envelope $0 \leq w \leq 2k_n$ this yields

$$|\varphi(O)| \leq \frac{2k_n}{\varepsilon},$$

so φ is square integrable under P_S . Applying the pull-out identity with $F(O) = H(O)D$ and $d = \Delta_D$, whose set-integral hypothesis is the receipt identity of [theorem 1](#),

$$\mathbb{E}_{P_S}[\varphi] = \mathbb{E}_{P_S^X}[w(X)\Delta_D(X)] = \mu_n,$$

and, bounding the variance by the second moment and using $|H(O)D| \leq 1/\varepsilon$,

$$\text{Var}_{P_S}(\varphi) \leq \mathbb{E}_{P_S}[\varphi^2] \leq \mathbb{E}_{P_S^X}\left[\frac{w(X)^2}{\varepsilon^2}\right] = \frac{\kappa_n}{\varepsilon^2}.$$

Since $\widehat{B}_n = n^{-1} \sum_{i \leq n} \varphi(O_i^S)$ averages n independent copies of φ ,

$$\mathbb{E}_P \widehat{B}_n = \mu_n, \quad \text{Var}_P(\widehat{B}_n) \leq \frac{\kappa_n}{\varepsilon^2 n}.$$

Chebyshev’s inequality at radius $\mu_n/2$ then gives, with $t_n = n\mu_n^2/\kappa_n$,

$$\Pr\left\{\frac{\mu_n}{2} < \left|\widehat{B}_n - \mu_n\right|\right\} \leq \frac{\kappa_n/(\varepsilon^2 n)}{(\mu_n/2)^2} = \frac{4}{\varepsilon^2 t_n}.$$

Likewise $\widehat{\kappa}_n = n^{-1} \sum_{i \leq n} w(X_i)^2 \geq 0$ averages n independent copies of $w(X)^2$, so it is integrable with

$$\mathbb{E}_P \widehat{\kappa}_n = \mathbb{E}_{P_S^X}[w(X)^2] = \kappa_n.$$

We now convert these moments into a length bound. Whenever $\widehat{B}_n \neq 0$, the score restriction $|\widehat{A}_n - \vartheta \widehat{B}_n| \leq L_\alpha \sqrt{\widehat{\kappa}_n/n}$ confines ϑ to the interval centred at $\widehat{A}_n/\widehat{B}_n$ of half-length $L_\alpha \sqrt{\widehat{\kappa}_n/n}/|\widehat{B}_n|$, while $C_n \subseteq \Theta$ and $\lambda(\Theta) = 2$ always; hence

$$\lambda(C_n) \leq \min \left\{ 2, \frac{2L_\alpha \sqrt{\widehat{\kappa}_n/n}}{|\widehat{B}_n|} \right\}.$$

On the event $\{|\widehat{B}_n - \mu_n| \leq \mu_n/2\}$ one has $|\widehat{B}_n| \geq \mu_n/2$, so there $\lambda(C_n) \leq 4L_\alpha \sqrt{\widehat{\kappa}_n/n}/\mu_n$; off that event use $\lambda(C_n) \leq 2$. Splitting the expectation accordingly and using concavity of the square root, $\mathbb{E}_P \sqrt{\widehat{\kappa}_n} \leq \sqrt{\mathbb{E}_P \widehat{\kappa}_n} = \sqrt{\kappa_n}$,

$$\mathbb{E}_P \lambda(C_n) \leq \frac{4L_\alpha}{\mu_n} \sqrt{\frac{\kappa_n}{n}} + 2 \Pr_P \left\{ \frac{\mu_n}{2} < |\widehat{B}_n - \mu_n| \right\} \leq 4L_\alpha t_n^{-1/2} + \frac{8}{\varepsilon^2 t_n},$$

where the first term was rewritten using $\sqrt{\kappa_n/n}/\mu_n = t_n^{-1/2}$. If $t_n \leq 1$ then $\min\{1, t_n^{-1/2}\} = 1$ and $\lambda(C_n) \leq 2 \leq C_0$ already gives the claim; if $t_n \geq 1$ then $t_n^{-1} \leq t_n^{-1/2}$, so the last display is at most $(4L_\alpha + 8/\varepsilon^2)t_n^{-1/2} \leq C_0 \min\{1, t_n^{-1/2}\}$. In both cases

$$\mathbb{E}_P \lambda(C_n) \leq C_0 \min\{1, t_n^{-1/2}\}.$$

4. **Pointwise coverage.** Define

$$\delta_n = \begin{cases} 1, & n = 0, \\ 16k_n^2/n, & n \geq 1. \end{cases}$$

The scale condition $k_n/\sqrt{n} \rightarrow 0$ gives $\delta_n \rightarrow 0$. At the index $n = 0$ the asserted bound reads $1 - \alpha - 1 = -\alpha \leq \Pr_P\{\theta_T \in C_n\}$, which holds because probabilities are nonnegative and $\alpha > 0$.

Fix $n \geq 1$ and $P \in \mathcal{P}_n$, retaining the notation of the previous item, and set

$$S_n = \widehat{A}_n - \theta_n \widehat{B}_n, \quad \psi_n(O) = w(X)H(O)\{Y - \theta_n D\}.$$

Expanding $\widehat{A}_n$ and $\widehat{B}_n$ and collecting terms observation by observation,

$$S_n = \frac{1}{n} \sum_{i=1}^n \psi_n(O_i^S).$$

Applying the pull-out identity twice, with $(F, d) = (H \cdot Y, \Delta_Y)$ and with $(F, d) = (H \cdot D, \Delta_D)$, whose set-integral hypotheses are exactly the two identities of [theorem 1](#),

$$\mathbb{E}_{P_S}[w(X)H(O)Y] = \mu_{Y,n}, \quad \mathbb{E}_{P_S}[w(X)H(O)D] = \mu_n,$$

so by the ratio identity $\mu_{Y,n} = \theta_n \mu_n$ of [theorem 1](#),

$$\mathbb{E}_{P_S}[\psi_n] = \mu_{Y,n} - \theta_n \mu_n = 0.$$

Since $Y \in [0, 1]$, $D \in \{0, 1\}$ and $\theta_n \in \Theta = [-1, 1]$, we have $|Y - \theta_n D| \leq 2$; with $|H(O)| \leq 1/\varepsilon$ from overlap this gives the pointwise bounds

$$|H(O)\{Y - \theta_n D\}| \leq \frac{2}{\varepsilon}, \quad \psi_n(O)^2 \leq \frac{4w(X)^2}{\varepsilon^2}.$$

Hence $\text{Var}_{P_S}(\psi_n) \leq \mathbb{E}_{P_S}[\psi_n^2] \leq 4\kappa_n/\varepsilon^2$, and averaging n independent copies,

$$\mathbb{E}_P S_n = 0, \quad \text{Var}_P(S_n) \leq \frac{4\kappa_n}{\varepsilon^2 n}.$$

For the empirical dispersion, the envelope $0 \leq w \leq 2\kappa_n$ gives the pointwise fourth-moment bound

$$w(x)^4 \leq 4k_n^2 w(x)^2,$$

so $\text{Var}_{P_S^X}(w(X)^2) \leq \mathbb{E}_{P_S^X}[w(X)^4] \leq 4k_n^2 \kappa_n$ and therefore

$$\text{Var}_P(\hat{\kappa}_n) \leq \frac{4k_n^2 \kappa_n}{n}, \quad \Pr_P \left\{ \hat{\kappa}_n < \frac{\kappa_n}{2} \right\} \leq \frac{4k_n^2 \kappa_n / n}{(\kappa_n/2)^2} = \frac{16k_n^2}{n\kappa_n} \leq \frac{16k_n^2}{n} = \delta_n,$$

the second inequality being Chebyshev's inequality at radius $\kappa_n/2$ and the last step using $\kappa_n \geq 1$.

With

$$a_n = L_\alpha \sqrt{\frac{\kappa_n}{2n}} > 0,$$

Chebyshev's inequality and the identity $L_\alpha^2 = 8/(\alpha\varepsilon^2)$ yield

$$\Pr_P \{|S_n| > a_n\} \leq \frac{\text{Var}_P(S_n)}{a_n^2} \leq \frac{4\kappa_n/(\varepsilon^2 n)}{L_\alpha^2 \kappa_n / (2n)} = \frac{8}{\varepsilon^2 L_\alpha^2} = \alpha.$$

Write $\mathcal{B}_n^\kappa = \{\hat{\kappa}_n < \kappa_n/2\}$ and $\mathcal{B}_n^S = \{|S_n| > a_n\}$. On the complement of $\mathcal{B}_n^\kappa \cup \mathcal{B}_n^S$ we have $\hat{\kappa}_n \geq \kappa_n/2$ and hence

$$|S_n| \leq a_n = L_\alpha \sqrt{\frac{\kappa_n}{2n}} \leq L_\alpha \sqrt{\frac{\hat{\kappa}_n}{n}},$$

which together with $\theta_n \in \Theta$ puts $\theta_n \in C_n$. Both bad events are functions of the source sample alone, and under the two-sample law the target covariate sample is independent of it, so the two-sample probability of the good event equals its source-sample probability. A union bound therefore gives

$$1 - \alpha - \delta_n \leq 1 - \Pr_P(\mathcal{B}_n^S) - \Pr_P(\mathcal{B}_n^\kappa) \leq \Pr_P \{\theta_T \in C_n\},$$

and this holds for every $P \in \mathcal{P}_n$.

5. **Oracle honesty.** Coverage probabilities lie in $[0, 1]$, and by the first item $\mathcal{P}_n \neq \emptyset$ for all sufficiently large n . On those indices the worst-case row of [definition 6](#), which is set to 1 when the class is empty, is the genuine infimum, and the previous item bounds it uniformly:

$$\inf_{P \in \mathcal{P}_n} \Pr_P \{\theta_T \in C_n\} \geq 1 - \alpha - \delta_n.$$

Since $\delta_n \rightarrow 0$, taking the liminf of these rows gives

$$1 - \alpha \leq \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{P}_n} \Pr_P \{\theta_T \in C_n\}.$$

Together with $0 < \alpha < 1$ this is exactly the requirement of [definition 6](#), so $(C_n) \in \mathfrak{C}^{\text{or}}$.

6. **Frontier risk of the constructed sequence.** Fix $t_0 > 0$. The map $t \mapsto \min\{1, t^{-1/2}\}$ is nonincreasing on $(0, \infty)$, so for every $n \geq 1$ and every $P \in \mathcal{P}_n$ with $t_n \geq t_0$ the pointwise bound of the third item gives

$$\mathbb{E}_P \lambda(C_n) \leq C_0 \min\{1, t_n^{-1/2}\} \leq C_0 \min\{1, t_0^{-1/2}\}.$$

The right-hand side is nonnegative and does not depend on P , so the row supremum obeys the same bound, including on rows where no such P exists and the supremum is 0; the single index $n = 0$ does not affect a limsup. Taking the limsup in [definition 7](#),

$$R((C_n), t_0) \leq C_0 \min\{1, t_0^{-1/2}\}.$$

7. **Oracle value.** Every reported set A satisfies $\lambda(A) \leq \lambda(\Theta) = 2$, so for every oracle honest $(E_n) \in \mathfrak{C}^{\text{or}}$ each row supremum in [definition 7](#) lies in $[0, 2]$ and hence

$$0 \leq R((E_n), t_0) \leq 2.$$

The infimum in [definition 8](#) is therefore over a set of reals bounded below by 0, and that set is nonempty because $(C_n) \in \mathfrak{C}^{\text{or}}$ by the fifth item. Evaluating the infimum at (C_n) and using the previous item,

$$V^*(t_0) \leq R((C_n), t_0) \leq C_0 \min\{1, t_0^{-1/2}\}.$$

For the reverse direction, take the admissible geometry $\mathfrak{g}$ of the first item. Its declared weight is a version of dP_T/dP_S^X for every law in the slice $\mathcal{P}_n(\mathfrak{g})$, so a globally oracle honest sequence, evaluated with that declared weight and propensity, is honest for the fixed geometry and its fixed-geometry risk is at most its global risk; consequently

$$V_{\mathfrak{g}}^*(t_0) \leq R((E_n), t_0) \quad \text{for every } (E_n) \in \mathfrak{C}^{\text{or}}.$$

By [theorem 4](#) applied at $\mathfrak{g}$, whose constants are $c_\alpha = 3(1 - \alpha)^2/16$ and $C_\alpha = C_0$,

$$\frac{3(1 - \alpha)^2}{16} \min\{1, t_0^{-1/2}\} \leq V_{\mathfrak{g}}^*(t_0) \leq C_0 \min\{1, t_0^{-1/2}\}.$$

Chaining the last two displays and taking the infimum over $(E_n) \in \mathfrak{C}^{\text{or}}$,

$$\frac{3(1 - \alpha)^2}{16} \min\{1, t_0^{-1/2}\} \leq V^*(t_0) \leq C_0 \min\{1, t_0^{-1/2}\}.$$

Together with the sixth item this establishes the asserted frontier-risk bound and the corresponding attainable oracle-value order at every fixed $t_0 > 0$. □

Proof of [theorem 7](#). Throughout, L is the fixed constant of the statement, so that

$$B_{\varepsilon, c} = 8(\varepsilon^{-2} + c^{-1}), \quad L \geq \left(\frac{2B_{\varepsilon, c}}{\alpha} \right)^{1/2}, \quad C_0 = \max \left\{ 2, 4\sqrt{2}L + 8B_{\varepsilon, c} \right\}.$$

Because $\varepsilon > 0$ and $c > 0$ we have $B_{\varepsilon, c} \geq 0$, hence $L \geq 0$, and since C_0 is a maximum with 2 as one of its arguments,

$$0 < 2 \leq C_0.$$

1. **Every row of the class is nonempty.** The carrier hypothesis supplies, for each n , distinct points $x_{1,n}, \dots, x_{k_n,n} \in \mathcal{X}$ whose singletons are measurable; write

$$\mathcal{X}_n = \{x_{1,n}, \dots, x_{k_n,n}\}$$

for the resulting realized support. Under $N_n/n \rightarrow c > 0$, $k_n \rightarrow \infty$, and $k_n/\sqrt{n} \rightarrow 0$, the uniform finite-cell array carried on $\mathcal{X}_n$, with source cell masses $1/k_n$ and propensity $1/2$, satisfies all conditions of $\mathcal{P}_n$ in [definition 3](#) together with [assumption 16](#). Since $0 < c_- \leq 1 \leq c_+$,

$$\frac{c_-}{k_n} \leq \frac{1}{k_n} \leq \frac{c_+}{k_n},$$

so this array also satisfies the cell-mass bounds of [definition 20](#) and therefore

$$\mathcal{N}_n^{\text{reg}} \neq \emptyset \quad \text{for every } n.$$

Consequently the empty-row convention in [definition 21](#) is never invoked, and the coverage requirement there reads

$$1 - \alpha \leq \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{N}_n^{\text{reg}}} P(\theta_T \in C_n^{\text{reg}}).$$

2. **The rule, and its almost-sure agreement with the displayed statistics.** The procedure receives at row n the source sample $(O_i^S)_{i=1}^n$, the target sample $(X_j^T)_{j=1}^{N_n}$, the realized support $\mathcal{X}_n$, and the two known k_n -vectors $(q_{x,n})_{x \in \mathcal{X}_n}$ and $(e_{x,n})_{x \in \mathcal{X}_n}$. It extends those two vectors off $\mathcal{X}_n$ by zero, i.e. it works with

$$\tilde{q}_n(x) = q_{x,n} \mathbf{1}\{x \in \mathcal{X}_n\}, \quad \tilde{e}_n(x) = e_{x,n} \mathbf{1}\{x \in \mathcal{X}_n\},$$

and returns the set displayed in the statement, namely Θ when $N_n < 2$ and otherwise

$$\left\{ \vartheta \in \Theta : \left| \widehat{M}_{Y,n} - \vartheta \widehat{M}_{D,n} \right| \leq L \sqrt{\widehat{K}_n/n} \right\},$$

with H_i , $\widehat{p}_{x,n}$, $\widehat{m}_{G,n}$, $\widehat{M}_{G,n}$, $\widehat{\kappa}_{U,n}$, and $\widehat{K}_n$ formed from $\tilde{q}_n$ and $\tilde{e}_n$ in place of $q_{x,n}$ and $e(\cdot)$. This rule uses only the source and target samples, the known cell masses, and the known propensity values, and its sample-by- ϑ graph on the realized support is measurable because $\mathcal{X}_n$ is a finite union of measurable singletons and the maps $s \mapsto \widehat{M}_{Y,n}$, $s \mapsto \widehat{M}_{D,n}$, $s \mapsto \widehat{K}_n$ are measurable. Hence it is a regular-cell procedure in the sense of [definition 21](#).

Fix n and $P \in \mathcal{N}_n^{\text{reg}}$. By [definition 20](#) the source covariate marginal is carried by the realized support, $P_S^X(\mathcal{X}_n) = 1$, and by [assumption 12](#) $P_T \ll P_S^X$, so also $P_T(\mathcal{X}_n) = 1$. Therefore

$$P\{X_i \in \mathcal{X}_n \text{ for all } i \leq n, \text{ and } X_j^T \in \mathcal{X}_n \text{ for all } j \leq N_n\} = 1,$$

and on that event $\tilde{q}_n$ and $\tilde{e}_n$ agree with $q_{x,n} = P_S^X(X = x)$ and $e(\cdot)$ at every realized covariate. Since $\widehat{m}_{G,n}$, $\widehat{M}_{G,n}$, and $\widehat{\kappa}_{U,n}$ depend on the mass and propensity functions only through their values at realized covariates, the set returned by the procedure coincides P -almost surely with the set built from $q_{x,n}$ and $e(\cdot)$:

$$P\{C_n^{\text{reg}} \text{ equals the rule of the statement evaluated at } q_{x,n} \text{ and } e\} = 1.$$

3. **Honesty.** Since $N_n/n \rightarrow c > 0$, there is an index beyond which $n \geq 1$ and $N_n \geq 2$; fix such an n and let $P \in \mathcal{N}_n^{\text{reg}}$. Write the centered inversion statistic at the true value,

$$\zeta_n = \widehat{M}_{Y,n} - \theta_T \widehat{M}_{D,n},$$

which is well defined because $\theta_T \in \Theta$ by [theorem 1](#). Three facts control ζ_n .

First, ζ_n is exactly centered:

$$\mathbb{E}_P[\zeta_n] = 0.$$

Indeed, the inverse-propensity weighting makes $\mathbb{E}_P[\widehat{m}_{G,n}(x)] = \Delta_G(x)$ for each cell x , the independent target sample gives $\mathbb{E}_P[\widehat{p}_{x,n}] = P_T(X = x) = q_{x,n}w(x)$, so $\mathbb{E}_P[\widehat{M}_{G,n}]$ equals the transported mean of Δ_G , and $\mu_{Y,n} - \theta_T \mu_n = 0$ by [theorem 1](#).

Second, uniformly over $P \in \mathcal{N}_n^{\text{reg}}$ for all large n ,

$$\text{Var}_P(\zeta_n) \leq \frac{B_{\varepsilon,c} \kappa_n}{n}, \quad \text{Var}_P(\widehat{M}_{D,n}) \leq \frac{B_{\varepsilon,c} \kappa_n}{n}, \quad \mathbb{E}_P[\widehat{M}_{D,n}] = \mu_n.$$

The two variance bounds hold because the cross-average splits into a source-sample part and a target-sample part: conditionally on the target sample the source part is an average of n independent terms whose inverse-propensity factor is bounded by ε^{-1} under [assumption 4](#) and whose inverse-mass weighting contributes the dispersion κ_n , giving at most $8\varepsilon^{-2}\kappa_n/n$; the target part is an average of N_n independent bounded terms, giving at most $4/N_n \leq 8/(cn) \leq 8c^{-1}\kappa_n/n$ once $N_n/n \rightarrow c > 0$ and $\kappa_n \geq 1$. The two contributions add to $B_{\varepsilon,c}\kappa_n/n$ with $B_{\varepsilon,c} = 8(\varepsilon^{-2} + c^{-1})$.

Third, the transport weight is a probability density with respect to P_S^X , so by Jensen's inequality its second moment satisfies

$$1 \leq \kappa_n.$$

Now decompose the noncoverage event. Off the full-measure support event of the [previous](#) step, and off the event $\{\widehat{K}_n \geq \kappa_n/2\}$, we discard; on the intersection of both, $L\sqrt{\widehat{K}_n/n} \geq L\sqrt{\kappa_n/(2n)}$, so $|\zeta_n| \leq L\sqrt{\kappa_n/(2n)}$ already forces $\theta_T \in C_n^{\text{reg}}$. Chebyshev's inequality applied with the variance bound gives

$$P\left\{|\zeta_n| > L\sqrt{\frac{\kappa_n}{2n}}\right\} \leq \frac{B_{\varepsilon,c}\kappa_n/n}{L^2\kappa_n/(2n)} = \frac{2B_{\varepsilon,c}}{L^2} \leq \alpha,$$

the last step because $L^2 \geq 2B_{\varepsilon,c}/\alpha$. For the remaining event put

$$\delta_n = \sup_{P \in \mathcal{N}_n^{\text{reg}}} P\left\{\widehat{K}_n < \kappa_n/2\right\}.$$

A Chebyshev bound for the target-sample U-statistic $\widehat{\kappa}_{U,n}$, whose variance is at most a constant multiple of k_n^2/N_n under the cell-mass lower bound $q_{x,n} \geq c_-/k_n$, gives

$$\delta_n \leq \frac{128 k_n^2}{c_-^2 N_n} \xrightarrow{n \rightarrow \infty} 0,$$

because $N_n/n \rightarrow c > 0$ and $k_n/\sqrt{n} \rightarrow 0$.

Combining the three events, for all large n and every $P \in \mathcal{N}_n^{\text{reg}}$,

$$P(\theta_T \in C_n^{\text{reg}}) \geq 1 - \alpha - \delta_n,$$

hence

$$1 - \alpha \leq \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{N}_n^{\text{reg}}} P(\theta_T \in C_n^{\text{reg}}),$$

and $(C_n^{\text{reg}}) \in \mathfrak{C}^{\text{reg}}$.

4. **Pointwise expected-length bound.** Fix $t_0 > 0$. Choose n_0 so that for every $n \geq n_0$ one has $n \geq 1$, $N_n \geq 2$, and, for every $P \in \mathcal{N}_n^{\text{reg}}$,

$$P\left\{\left|\widehat{M}_{D,n} - \mu_n\right| > \frac{\mu_n}{2}\right\} \leq \frac{4B_{\varepsilon,c}}{t_n}, \quad t_n = \frac{n\mu_n^2}{\kappa_n};$$

this is Chebyshev's inequality applied to $\widehat{M}_{D,n}$ at radius $\mu_n/2$, using the mean and variance recorded in the previous step, since $(B_{\varepsilon,c}\kappa_n/n)/(\mu_n/2)^2 = 4B_{\varepsilon,c}/t_n$.

Fix $n \geq n_0$ and $P \in \mathcal{N}_n^{\text{reg}}$. By [theorem 1](#) the transported first stage equals the target complier share,

$$\mu_n = P_n^F\{D(1) = 1, D(0) = 0 \mid S = 0\},$$

which is positive by [assumption 11](#); thus $\mu_n > 0$. The collision terms are nonnegative because $q_{x,n} > 0$ on $\mathcal{X}_n$, so

$$\widehat{\kappa}_{U,n} \geq 0, \quad \widehat{K}_n = 1 + \widehat{\kappa}_{U,n} \geq 1,$$

and $\widehat{K}_n$ is integrable under the two-sample law. Its mean is

$$\mathbb{E}_P[\widehat{K}_n] = 1 + \kappa_n \leq 2\kappa_n,$$

the identity because $\widehat{\kappa}_{U,n}$ is an unbiased U-statistic for κ_n over the $N_n(N_n - 1)$ ordered distinct target pairs, and the inequality because $1 \leq \kappa_n$.

By the previous step, C_n^{reg} agrees almost surely with the inversion of one affine inequality intersected with Θ . If $\widehat{M}_{D,n} \neq 0$, that inversion is the interval of centre $\widehat{M}_{Y,n}/\widehat{M}_{D,n}$ and half-length $L\sqrt{\widehat{K}_n/n}/|\widehat{M}_{D,n}|$, so

$$\lambda(C_n^{\text{reg}}) \leq \min\left\{2, \frac{2L\sqrt{\widehat{K}_n/n}}{|\widehat{M}_{D,n}|}\right\},$$

and in all cases $\lambda(C_n^{\text{reg}}) \leq \lambda(\Theta) = 2$.

On the event $\{|\widehat{M}_{D,n} - \mu_n| \leq \mu_n/2\}$ we have $|\widehat{M}_{D,n}| \geq \mu_n/2$, hence

$$\lambda(C_n^{\text{reg}}) \leq \frac{4L}{\mu_n} \sqrt{\frac{\widehat{K}_n}{n}},$$

while on the complementary event we use $\lambda(C_n^{\text{reg}}) \leq 2$. Taking expectations and bounding the complementary event by the displayed deviation probability,

$$\mathbb{E}_P[\lambda(C_n^{\text{reg}})] \leq \frac{4L}{\mu_n} \mathbb{E}_P \left[\sqrt{\frac{\widehat{K}_n}{n}} \right] + 2 \cdot \frac{4B_{\varepsilon,c}}{t_n} \leq \frac{4L}{\mu_n} \sqrt{\frac{\mathbb{E}_P[\widehat{K}_n]}{n}} + \frac{8B_{\varepsilon,c}}{t_n},$$

the second inequality by concavity of the square root. Inserting $\mathbb{E}_P[\widehat{K}_n] \leq 2\kappa_n$ and $t_n = n\mu_n^2/\kappa_n$,

$$\frac{4L}{\mu_n} \sqrt{\frac{2\kappa_n}{n}} = 4\sqrt{2} L t_n^{-1/2}, \quad \text{so} \quad \mathbb{E}_P[\lambda(C_n^{\text{reg}})] \leq \min \left\{ 2, 4\sqrt{2} L t_n^{-1/2} + \frac{8B_{\varepsilon,c}}{t_n} \right\}.$$

Finally, for $t > 0$, $u \geq 0$, and $v \geq 0$,

$$\min \left\{ 2, u t^{-1/2} + v t^{-1} \right\} \leq \max \{ 2, u + v \} \min \{ 1, t^{-1/2} \} :$$

for $t \leq 1$ the left side is at most $2 \leq \max \{ 2, u + v \} = \max \{ 2, u + v \} \min \{ 1, t^{-1/2} \}$, and for $t \geq 1$ one has $t^{-1} \leq t^{-1/2}$, so the left side is at most $(u + v)t^{-1/2}$. With $u = 4\sqrt{2} L$, $v = 8B_{\varepsilon,c}$, and $t = t_n$ this gives, for every $n \geq n_0$ and every $P \in \mathcal{N}_n^{\text{reg}}$,

$$\mathbb{E}_P[\lambda(C_n^{\text{reg}})] \leq C_0 \min \left\{ 1, t_n^{-1/2} \right\}.$$

5. **The frontier upper bound.** Expected lengths are nonnegative and the rows $n < n_0$ do not affect a limsup. Since $t \mapsto \min \{ 1, t^{-1/2} \}$ is nonincreasing on $(0, \infty)$, the pointwise bound of the previous step gives, for every $n \geq n_0$,

$$\sup_{P \in \mathcal{N}_n^{\text{reg}}: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n^{\text{reg}})] \leq C_0 \min \left\{ 1, t_0^{-1/2} \right\},$$

and therefore

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n^{\text{reg}}: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n^{\text{reg}})] \leq C_0 \min \left\{ 1, t_0^{-1/2} \right\} = \max \left\{ 2, 4\sqrt{2} L + 8B_{\varepsilon,c} \right\} \min \left\{ 1, t_0^{-1/2} \right\}.$$

This is the asserted upper bound.

6. **A deterministic geometry inside the regular-cell class.** Fix $t_0 > 0$. Using the points $x_{1,n}, \dots, x_{k_n,n}$ of the first step, define the deterministic array $\mathbf{g} = (\nu_n, w_n, e_n)_{n \geq 1}$ by

$$\nu_n = \sum_{i=1}^{k_n} \frac{1}{k_n} \delta_{x_{i,n}}, \quad w_n \equiv 1, \quad e_n \equiv \frac{1}{2},$$

with both the source covariate law and the target covariate law taken equal to ν_n . Since $k_n \cdot k_n^{-1} = 1$, each ν_n is a probability measure, and

$$\nu_n(\mathcal{X}_n) = 1, \quad \nu_n(\{x_{i,n}\}) = \frac{1}{k_n} \quad \text{for every } i.$$

The four clauses of [definition 15](#) hold:

$$\varepsilon \leq \frac{1}{2} \leq 1 - \varepsilon \quad (\text{from } 0 < \varepsilon < 1/2), \quad 0 \leq 1 \leq 2k_n \quad (\text{from } k_n \geq 1),$$

$$\int w_n d\nu_n = 1, \quad \kappa_n = \int w_n^2 d\nu_n = 1 \leq k_n, \quad \nu_n(A) = \int_A w_n d\nu_n \text{ for every measurable } A.$$

Hence $\mathbf{g} \in \mathcal{G}$. Moreover, since $0 < c_- \leq 1 \leq c_+$ gives $c_-/k_n \leq 1/k_n \leq c_+/k_n$, every law in the fixed-geometry slice $\mathcal{P}_n(\mathbf{g})$ of [definition 16](#) has source covariate marginal ν_n , which is carried by the k_n measurable singletons of $\mathcal{X}_n$ with masses in $[c_-/k_n, c_+/k_n]$; therefore

$$\mathcal{P}_n(\mathbf{g}) \subseteq \mathcal{N}_n^{\text{reg}} \quad \text{for every } n.$$

Through this inclusion the model restrictions assumed in the statement supply [assumptions 3, 4](#) and [15](#) on every slice row, and [assumptions 13](#) and [14](#) hold there because slice rows belong to $\mathcal{P}_n$. [Theorem 4](#) therefore applies and gives

$$\frac{3(1-\alpha)^2}{16} \min \left\{ 1, t_0^{-1/2} \right\} \leq V_{\mathbf{g}}^*(t_0).$$

We also record, from the least-favorable construction of [definition 19](#) at this geometry, that the slices are eventually inhabited at strength t_0 :

$$\text{for all large } n, \quad \{P \in \mathcal{P}_n(\mathbf{g}) : t_n \geq t_0\} \neq \emptyset.$$

7. Domination of the fixed-geometry problem by the regular-cell problem. Let $C = (C_n) \in \mathfrak{C}^{\text{reg}}$ be arbitrary. Define a fixed-geometry oracle procedure $D = (D_n)$ by

$$D_n \left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w, e \right) = \begin{cases} C_n \left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, \mathcal{X}_n, (1/k_n)_{x \in \mathcal{X}_n}, (e(x))_{x \in \mathcal{X}_n} \right), & E_n \text{ holds,} \\ \Theta, & \text{otherwise,} \end{cases}$$

where

$$E_n = \{X_i \in \mathcal{X}_n \text{ for all } i \leq n, \text{ and } X_j^T \in \mathcal{X}_n \text{ for all } j \leq N_n\}.$$

Then $D_n \subseteq \Theta$ because $C_n \subseteq \Theta$; the event E_n is measurable, being a finite intersection of preimages of the measurable set $\mathcal{X}_n$, and the sample-by- ϑ graph of D_n is the union of the graph of C_n restricted to E_n with $E_n^c \times \Theta$, hence measurable by the graph-measurability clause of [definition 21](#). Finally, D_n does not read the weight input w at all, so it is trivially invariant across versions of dP_T/dP_S^X . Thus D is an admissible oracle procedure for the class in [definition 17](#).

Let n be arbitrary and $P \in \mathcal{P}_n(\mathbf{g})$. Both the source and the target covariate law of P equal ν_n , and $\nu_n(\mathcal{X}_n) = 1$, so

$$P(E_n) = 1.$$

On E_n the arguments passed to C_n by D_n are exactly the class-supplied ones, because the source cell masses of P are $q_{x,n} = \nu_n(\{x\}) = 1/k_n$ on $\mathcal{X}_n$ and the propensity of P is $e_n \equiv 1/2$. Hence D_n and C_n return the same set P -almost surely, and therefore, for every n and every $P \in \mathcal{P}_n(\mathbf{g})$,

$$P(\theta_T \in D_n) = P(\theta_T \in C_n), \quad \mathbb{E}_P[\lambda(D_n)] = \mathbb{E}_P[\lambda(C_n)].$$

Because $\mathcal{P}_n(\mathfrak{g}) \subseteq \mathcal{N}_n^{\text{reg}}$ and both index sets are nonempty for all large n by the previous step, taking infima over the two classes and using the displayed coverage identity gives, for all large n ,

$$\inf_{P \in \mathcal{N}_n^{\text{reg}}} P(\theta_T \in C_n) \leq \inf_{P \in \mathcal{P}_n(\mathfrak{g})} P(\theta_T \in D_n).$$

Coverages lie in $[0, 1]$, so both sequences are bounded and the $\liminf$ comparison is legitimate; combined with $C \in \mathfrak{C}^{\text{reg}}$ this yields

$$1 - \alpha \leq \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{N}_n^{\text{reg}}} P(\theta_T \in C_n) \leq \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{P}_n(\mathfrak{g})} P(\theta_T \in D_n),$$

so $D \in \mathfrak{C}^{\text{or}}(\mathfrak{g})$.

The same almost-sure equality gives the length comparison. Every reported set lies in Θ , so all expected lengths lie in $[0, \lambda(\Theta)] = [0, 2]$; using the inclusion $\{P \in \mathcal{P}_n(\mathfrak{g}) : t_n \geq t_0\} \subseteq \{P \in \mathcal{N}_n^{\text{reg}} : t_n \geq t_0\}$ rowwise and the boundedness for the $\limsup$ comparison,

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}_n(\mathfrak{g}) : t_n \geq t_0} \mathbb{E}_P[\lambda(D_n)] \leq \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n^{\text{reg}} : t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)].$$

Fixed-geometry risks are nonnegative, so the infimum defining $V_{\mathfrak{g}}^*(t_0)$ in [definition 18](#) is bounded below and, D being admissible for it,

$$V_{\mathfrak{g}}^*(t_0) \leq \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}_n(\mathfrak{g}) : t_n \geq t_0} \mathbb{E}_P[\lambda(D_n)] \leq \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n^{\text{reg}} : t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)].$$

Chaining with the lower bound of the previous step and taking the infimum over $C \in \mathfrak{C}^{\text{reg}}$ gives

$$\frac{3(1 - \alpha)^2}{16} \min \left\{ 1, t_0^{-1/2} \right\} \leq \inf_{C \in \mathfrak{C}^{\text{reg}}} \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n^{\text{reg}} : t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)].$$

Together with $C_0 > 0$, the almost-sure representation, the membership $(C_n^{\text{reg}}) \in \mathfrak{C}^{\text{reg}}$, the upper bound of the fifth step, and this lower bound, the stated regular-cell attainment conclusion follows for every $t_0 > 0$. $\square$

Proof of [theorem 6](#). Throughout, $\lambda(A)$ is the Lebesgue measure of $A \cap \Theta$ as in [definition 7](#), and rows over an empty family are read with the standard conventions: a rowwise coverage infimum is evaluated as 1, and a rowwise strength-restricted expected-length supremum as 0. The coverage convention is the one [definition 21](#) records for $\mathcal{N}_n^{\text{reg}}$ and [definition 6](#) records for $\mathcal{P}_n$; the same reading is used below for the coverage rows over $\mathcal{N}_n$ and over $\mathcal{P}_n(\mathfrak{g})$, and the length convention for every strength-restricted row.

1. Fix the constants of the statement,

$$B_c = 32(1 + c^{-1}), \quad L_{\alpha, c} = \left(\frac{2B_c}{\alpha} \right)^{1/2}, \quad C_0 = \max\{2, 4\sqrt{2}L_{\alpha, c} + 8B_c\},$$

so that $C_0 \geq 2 > 0$. This is the first assertion.

Next, every finite-cell row class is inhabited:

$$\mathcal{N}_n \neq \emptyset \quad \text{for every } n.$$

An explicit witness is available: put the uniform law on k_n injected measurable singletons of the ambient carrier, take the target covariate law equal to it, transport weight $w \equiv 1$, source propensity $e \equiv 1/2$, and conditional complier probability $1/(n+1)$ at row n ; under $c > 0$ with $N_n/n \rightarrow c$, $k_n > 0$ with $k_n \rightarrow \infty$ and $k_n/\sqrt{n} \rightarrow 0$, and $0 < \varepsilon < 1/2$, this array satisfies every clause of [definition 22](#).

Finally we calibrate an auxiliary overlap level at which the regular-cell result will be invoked. Set

$$\gamma = 4 + \frac{3}{c}, \quad a = \gamma^{-1/2}, \quad \bar{\varepsilon} = \frac{1}{4} + \frac{a}{2}.$$

Since $c > 0$ we have $\gamma > 4$ and hence $\sqrt{\gamma} > 2$, so

$$0 < a = \frac{1}{\sqrt{\gamma}} < \frac{1}{2}, \quad 0 < \frac{1}{4} < \bar{\varepsilon} < \frac{1}{2}.$$

Moreover $a\sqrt{\gamma} = 1$, so

$$\bar{\varepsilon}\sqrt{\gamma} = \frac{\sqrt{\gamma}}{4} + \frac{a\sqrt{\gamma}}{2} = \frac{\sqrt{\gamma}}{4} + \frac{1}{2} > \frac{1}{2} + \frac{1}{2} = 1.$$

Dividing by $\bar{\varepsilon} > 0$ gives $\bar{\varepsilon}^{-1} \leq \sqrt{\gamma}$, and squaring gives

$$\bar{\varepsilon}^{-2} \leq \gamma = 4 + \frac{3}{c}.$$

Write $B_{\bar{\varepsilon},c}$ for the regular-cell variance constant of [theorem 7](#) evaluated at overlap level $\bar{\varepsilon}$,

$$B_{\bar{\varepsilon},c} = 8(\bar{\varepsilon}^{-2} + c^{-1}).$$

Then $B_{\bar{\varepsilon},c} \geq 0$, and the previous display gives the calibration that fixes γ :

$$B_{\bar{\varepsilon},c} \leq 8\left(4 + \frac{3}{c} + \frac{1}{c}\right) = 32 + \frac{32}{c} = B_c.$$

Since $\alpha > 0$ and $t \mapsto t^{1/2}$ is increasing,

$$\left(\frac{2B_{\bar{\varepsilon},c}}{\alpha}\right)^{1/2} \leq \left(\frac{2B_c}{\alpha}\right)^{1/2} = L_{\alpha,c}.$$

2. Apply [theorem 7](#) with sampling ratio c , noncoverage level α , overlap constant $\bar{\varepsilon}$, regular-cell mass constants $c_- = c_+ = 1$, the same sequences N_n and k_n , the same measurable carrier, and critical constant $L_{\alpha,c}$. Write $\mathcal{N}_n^{\text{reg}}$ for the regular finite-cell class of [definition 20](#) formed with these constants, and $\mathfrak{C}^{\text{reg}}$ for the corresponding feasible class of [definition 21](#). Its hypotheses hold: $c > 0$ and $N_n/n \rightarrow c$; $0 < \bar{\varepsilon} < 1/2$ by Step 1 and $0 < \alpha < 1$; $0 < c_- = 1 \leq 1 \leq c_+ = 1$; $k_n > 0$, $k_n \rightarrow \infty$ and $k_n/\sqrt{n} \rightarrow 0$; the ambient-carrier condition; the two-sample, instrument-overlap and degrading-array restrictions, each of which is one of the defining conditions of $\mathcal{N}_n^{\text{reg}}$, since [definition 20](#) places $\mathcal{N}_n^{\text{reg}}$ inside $\mathcal{P}_n$ of [definition 3](#); and the critical-value requirement $(2B_{\bar{\varepsilon},c}/\alpha)^{1/2} \leq L_{\alpha,c}$ established in Step 1. The conclusion supplies a regular-cell procedure $C_n^{\text{reg}} = (C_n^{\text{reg}})$ with three properties. First, for every n and every $P \in \mathcal{N}_n^{\text{reg}}$, P -almost surely

$$C_n^{\text{reg}} = \left\{ \vartheta \in \Theta : \left| \widehat{M}_{Y,n} - \vartheta \widehat{M}_{D,n} \right| \leq L_{\alpha,c} \sqrt{\widehat{K}_n/n} \right\} \quad (N_n \geq 2), \quad C_n^{\text{reg}} = \Theta \quad (N_n < 2),$$

where $\widehat{M}_{Y,n}$, $\widehat{M}_{D,n}$ and $\widehat{K}_n$ are built from the law's own source-cell masses $q_{x,n} = P_S^X(X = x)$ and its own propensity e exactly as displayed in [theorem 7](#); call this set $C_n^\dagger$. Second, $C_n^{\text{reg}} \in \mathfrak{C}^{\text{reg}}$. Third, for every $t_0 > 0$,

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n^{\text{reg}}: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n^{\text{reg}})] \leq \max\{2, 4\sqrt{2}L_{\alpha,c} + 8B_{\bar{\varepsilon},c}\} \min\{1, t_0^{-1/2}\}.$$

3. Transport this regular-cell rule to the finite-cell experiment. First, every uniform finite-cell law is a regular finite-cell law with unit mass constants:

$$\mathcal{N}_n \subseteq \mathcal{N}_n^{\text{reg}} \quad \text{for every } n.$$

Indeed, let $P \in \mathcal{N}_n$. By [assumption 16](#) its source covariate marginal is supported on exactly k_n measurable cells with

$$q_{x,n} = \frac{1}{k_n},$$

which meets the two-sided mass bound $c_-/k_n \leq q_{x,n} \leq c_+/k_n$ of [definition 20](#) with $c_- = c_+ = 1$ as an equality; and $e(X) = 1/2$ P_S -almost surely, so that $0 < \bar{\varepsilon} < 1/2$ gives the overlap requirement at level $\bar{\varepsilon}$,

$$\bar{\varepsilon} \leq e(X) = \frac{1}{2} \leq 1 - \bar{\varepsilon} \quad P_S\text{-a.s.}$$

The remaining conditions of [definition 20](#) are those of [definition 3](#), which $\mathcal{N}_n$ inherits by [definition 22](#).

Let $C = (C_n)$ be the sample-only rule displayed in the statement: for $N_n \geq 2$,

$$C_n = \left\{ \vartheta \in \Theta : \left| \widehat{M}_{Y,n} - \vartheta \widehat{M}_{D,n} \right| \leq L_{\alpha,c} \sqrt{\widehat{K}_n/n} \right\}, \quad C_n = \Theta \quad (N_n < 2),$$

with $H_i = 2(2Z_i - 1)$, $\widehat{p}_{x,n}$, $\widehat{m}_{G,n}$, $\widehat{M}_{G,n}$, $\widehat{\kappa}_{U,n}$ and $\widehat{K}_n = 1 + \widehat{\kappa}_{U,n}$ as in the statement. This is exactly the regular-cell inversion rule of Step 2 run with the constant inputs

$$q_{x,n} \equiv \frac{1}{k_n}, \quad e \equiv \frac{1}{2},$$

so it uses only $(O_i^S)_{i=1}^n$ and $(X_j^T)_{j=1}^{N_n}$, takes values in Θ , and has a measurable sample-by- ϑ graph on the realized cell experiment. In particular the almost-sure agreement with the finite-cell score-inversion rule asserted in the statement holds by construction.

Now fix n and $P \in \mathcal{N}_n$. Three almost-sure identities combine. (i) P -almost surely every sampled source covariate X_i and every sampled target covariate X_j^T lies in the k_n cells of P_S^X , and on those points the constant inputs $1/k_n$ and $1/2$ coincide with the law's own $q_{x,n}$ and e ; hence P -almost surely C_n equals the regular-cell inversion rule fed P 's own cell design, masses and propensity. (ii) That rule, evaluated on the law's own finite cell vectors, agrees P -almost surely with $C_n^\dagger$, because the sampled covariates almost surely lie in the realized support where the finite vectors and their ambient extensions coincide. (iii) $C_n^{\text{reg}} = C_n^\dagger$ P -almost surely by Step 2. Chaining (i)–(iii),

$$C_n = C_n^{\text{reg}} \quad P\text{-almost surely, for every } n \text{ and every } P \in \mathcal{N}_n,$$

and therefore

$$P(\theta_T \in C_n) = P(\theta_T \in C_n^{\text{reg}}), \quad \mathbb{E}_P[\lambda(C_n)] = \mathbb{E}_P[\lambda(C_n^{\text{reg}})].$$

Two consequences follow. For coverage: by the inclusion above, both row classes are nonempty for every n by Step 1, and an infimum over the larger class is no larger, so

$$\inf_{P \in \mathcal{N}_n^{\text{reg}}} P(\theta_T \in C_n^{\text{reg}}) \leq \inf_{P \in \mathcal{N}_n} P(\theta_T \in C_n) \quad \text{for every } n.$$

Coverage probabilities lie in $[0, 1]$, so both row sequences are bounded and the inequality passes through $\liminf$; with $C^{\text{reg}} \in \mathfrak{C}^{\text{reg}}$ from Step 2 this gives

$$1 - \alpha \leq \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{N}_n^{\text{reg}}} P(\theta_T \in C_n^{\text{reg}}) \leq \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{N}_n} P(\theta_T \in C_n),$$

that is, $C \in \mathfrak{C}^{\text{cell}}$ in the sense of [definition 10](#).

For risk: fix $t_0 > 0$. The inclusion also preserves the strength restriction $t_n \geq t_0$, so for every n

$$\sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)] \leq \sup_{P \in \mathcal{N}_n^{\text{reg}}: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n^{\text{reg}})].$$

Every expected length is nonnegative and, by $\lambda(A) \leq \lambda(\Theta) = 2$, at most 2; hence both row sequences lie in $[0, 2]$ and the inequality passes through $\limsup$:

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)] \leq \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n^{\text{reg}}: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n^{\text{reg}})].$$

4. Fix $t_0 > 0$. Chaining the risk comparison of Step 3, the regular-cell bound of Step 2, and the constant comparison $B_{\bar{\varepsilon}, c} \leq B_c$ of Step 1 (legitimate because $\min\{1, t_0^{-1/2}\} \geq 0$),

$$\begin{aligned} \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)] &\leq \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n^{\text{reg}}: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n^{\text{reg}})] \\ &\leq \max\{2, 4\sqrt{2}L_{\alpha, c} + 8B_{\bar{\varepsilon}, c}\} \min\{1, t_0^{-1/2}\} \\ &\leq \max\{2, 4\sqrt{2}L_{\alpha, c} + 8B_c\} \min\{1, t_0^{-1/2}\} \\ &= C_0 \min\{1, t_0^{-1/2}\}. \end{aligned}$$

This is the asserted attainment bound, for every $t_0 > 0$.

5. For the converse, fix $t_0 > 0$. For each n choose an injection $i \mapsto x_{i,n}$ of $\{1, \dots, k_n\}$ into the measurable carrier whose singletons $\{x_{i,n}\}$ are measurable, and write

$$\mathcal{X}_n = \{x_{1,n}, \dots, x_{k_n,n}\}.$$

Define the deterministic geometry $\mathfrak{g} = (P_{S,n}^X, w, e)_{n \geq 1}$ by

$$P_{S,n}^X = P_{T,n} = \sum_{i=1}^{k_n} k_n^{-1} \delta_{x_{i,n}}, \quad w \equiv 1, \quad e \equiv \frac{1}{2}.$$

Then $\mathbf{g} \in \mathcal{G}$ in the sense of [definition 15](#). Both covariate laws are probability measures because $k_n \cdot k_n^{-1} = 1$. The overlap clause holds because $0 < \varepsilon < 1/2$ gives

$$\varepsilon \leq e(x) = \frac{1}{2} \leq 1 - \varepsilon.$$

The envelope clause holds because $k_n \geq 1$ gives

$$0 \leq w(x) = 1 \leq 2k_n.$$

The normalization and second-moment clauses are

$$\mathbb{E}_S[w(X)] = 1, \quad \kappa_n = \mathbb{E}_S[w(X)^2] = 1 \leq k_n,$$

and the target-law clause holds because, for every measurable $A \subseteq \mathcal{X}$,

$$\mathbb{E}_S[w(X)\mathbf{1}\{X \in A\}] = P_{S,n}^X(A) = P_{T,n}(A).$$

6. The selected range is measurable, being the finite union $\mathcal{X}_n = \bigcup_{i=1}^{k_n} \{x_{i,n}\}$ of measurable singletons, and

$$P_{S,n}^X(\mathcal{X}_n) = P_{T,n}(\mathcal{X}_n) = 1, \quad P_{S,n}^X(\{x_{i,n}\}) = k_n^{-1} \quad (1 \leq i \leq k_n).$$

Hence every $P \in \mathcal{P}_n(\mathbf{g})$ satisfies [assumption 16](#) with these cells: by [definition 16](#) its source covariate marginal is exactly $P_{S,n}^X$, which is supported on the k_n cells and assigns each mass $1/k_n$, and its propensity is $e \equiv 1/2$. All remaining conditions of [definition 22](#) are the conditions of [definition 3](#), which hold because $\mathcal{P}_n(\mathbf{g}) \subseteq \mathcal{P}_n$. Therefore

$$\mathcal{P}_n(\mathbf{g}) \subseteq \mathcal{N}_n \quad \text{for every } n.$$

The same measurable range also makes

$$E_n = \bigcap_{i=1}^n \{X_i \in \mathcal{X}_n\} \cap \bigcap_{j=1}^{N_n} \{X_j^T \in \mathcal{X}_n\}$$

a measurable event of the two-sample space, being a finite intersection of preimages of $\mathcal{X}_n$ under coordinate projections.

7. [Theorem 4](#) applies to $\mathbf{g}$: its target-sample-ratio, envelope-scale, overlap and noncoverage hypotheses are the standing hypotheses of the present statement, its geometry hypothesis is Step 5, and its fixed-geometry-slice hypothesis holds because $\mathcal{P}_n(\mathbf{g}) \subseteq \mathcal{P}_n$ and the five conditions it lists are among those of [definition 3](#). Its lower half gives, with $c_\alpha = 3(1 - \alpha)^2/16$,

$$\frac{3(1 - \alpha)^2}{16} \min\{1, t_0^{-1/2}\} \leq V_{\mathbf{g}}^*(t_0).$$

It therefore suffices to show that for every feasible finite-cell honest $\tilde{C} \in \mathfrak{C}^{\text{cell}}$,

$$V_{\mathbf{g}}^*(t_0) \leq \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(\tilde{C}_n)];$$

taking the infimum over $\tilde{C}$ then yields the asserted converse. That infimum is over a nonempty class, since $C \in \mathfrak{C}^{\text{cell}}$ by Step 3.

8. Fix $\tilde{C} \in \mathfrak{C}^{\text{cell}}$ and define a fixed-geometry oracle procedure $D = (D_n)$ by

$$D_n = \begin{cases} \tilde{C}_n, & \text{on } E_n, \\ \Theta, & \text{on } E_n^c. \end{cases}$$

Then $D_n \subseteq \Theta$. Its sample-by- ϑ graph is

$$\left\{ (s, \vartheta) : s \in E_n, \vartheta \in \tilde{C}_n(s) \right\} \cup (E_n^c \times \Theta),$$

which is measurable: the first set is the restriction to the measurable event E_n of the graph of $\tilde{C}_n$, which is measurable on the realized cell experiment with cells $\mathcal{X}_n$, and the second is a product of measurable sets. Finally D_n is a function of the samples alone, so it does not change when the oracle density-ratio input is replaced by another version, which is the weight-invariance requirement imposed on oracle procedures in [definition 6](#).

9. Let $P \in \mathcal{P}_n(\mathfrak{g})$. Its source and target covariate laws are both $P_{S,n}^X = P_{T,n}$, which gives $\mathcal{X}_n$ probability one by Step 6, and the two samples have independent coordinates, so

$$P(E_n) = 1.$$

Hence $D_n = \tilde{C}_n$ P -almost surely, and therefore

$$P(\theta_T \in D_n) = P(\theta_T \in \tilde{C}_n), \quad \mathbb{E}_P[\lambda(D_n)] = \mathbb{E}_P[\lambda(\tilde{C}_n)].$$

For all sufficiently large n the fixed-geometry threshold slice $\{P \in \mathcal{P}_n(\mathfrak{g}) : t_n \geq t_0\}$ is inhabited, hence so is $\mathcal{P}_n(\mathfrak{g})$; and $\mathcal{N}_n \neq \emptyset$ for every n by Step 1. For such n , the inclusion of Step 6 and the coverage equality just displayed give

$$\inf_{P \in \mathcal{N}_n} P(\theta_T \in \tilde{C}_n) \leq \inf_{P \in \mathcal{P}_n(\mathfrak{g})} P(\theta_T \in D_n).$$

The left-hand row sequence is bounded below by 0 and the right-hand one is eventually bounded above by 1, since coverage probabilities lie in $[0, 1]$; so $\liminf$ preserves this eventual inequality, and honesty of $\tilde{C}$ over $\mathcal{N}_n$ gives

$$1 - \alpha \leq \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{N}_n} P(\theta_T \in \tilde{C}_n) \leq \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{P}_n(\mathfrak{g})} P(\theta_T \in D_n),$$

that is, $D \in \mathfrak{C}^{\text{or}}(\mathfrak{g})$ in the sense of [definition 17](#).

10. Every confidence set is measured inside the compact parameter space, so

$$\lambda(A) \leq \lambda(\Theta) = 2 \quad \text{for every } A \subseteq \mathbb{R}.$$

Consequently $\mathbb{E}_P[\lambda(\tilde{C}_n)] \leq 2$ for every n and every $P \in \mathcal{N}_n$, and $\mathbb{E}_P[\lambda(E'_n)] \leq 2$ for every fixed-geometry oracle procedure (E'_n) and every $P \in \mathcal{P}_n(\mathfrak{g})$; all these expectations are nonnegative.

For all sufficiently large n the fixed-geometry threshold slice is inhabited; on those n , the inclusion of Step 6 (which also preserves the restriction $t_n \geq t_0$) and the equality of expected lengths from Step 9 give

$$\sup_{P \in \mathcal{P}_n(\mathfrak{g}) : t_n \geq t_0} \mathbb{E}_P[\lambda(D_n)] \leq \sup_{P \in \mathcal{N}_n : t_n \geq t_0} \mathbb{E}_P[\lambda(\tilde{C}_n)].$$

The left-hand row sequence is eventually nonnegative, because the slice is eventually inhabited and expected lengths are nonnegative, and the right-hand row sequence is bounded above by 2 for every n by the length bound (with the empty-row supremum evaluated as 0). These two bounds let the eventual inequality pass through $\limsup$:

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}_n(\mathfrak{g}): t_n \geq t_0} \mathbb{E}_P[\lambda(D_n)] \leq \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(\tilde{C}_n)].$$

11. It remains to place the fixed-geometry value below the risk of this particular D . By the length bound of Step 10, for every fixed-geometry oracle procedure (E'_n) each row supremum lies in $[0, 2]$, so

$$0 \leq \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}_n(\mathfrak{g}): t_n \geq t_0} \mathbb{E}_P[\lambda(E'_n)].$$

The set of fixed-geometry risks over $\mathfrak{C}^{\text{or}}(\mathfrak{g})$ is therefore bounded below by 0, and it contains the risk of D because $D \in \mathfrak{C}^{\text{or}}(\mathfrak{g})$ by Step 9. Hence the infimum defining $V_{\mathfrak{g}}^*(t_0)$ in [definition 18](#) satisfies

$$V_{\mathfrak{g}}^*(t_0) \leq \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}_n(\mathfrak{g}): t_n \geq t_0} \mathbb{E}_P[\lambda(D_n)].$$

Combining with Step 10,

$$V_{\mathfrak{g}}^*(t_0) \leq \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(\tilde{C}_n)].$$

Since $\tilde{C} \in \mathfrak{C}^{\text{cell}}$ was arbitrary, taking the infimum over that nonempty class and combining with Step 7 yields

$$\frac{3(1-\alpha)^2}{16} \min\{1, t_0^{-1/2}\} \leq V_{\mathfrak{g}}^*(t_0) \leq \inf_{\tilde{C} \in \mathfrak{C}^{\text{cell}}} \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(\tilde{C}_n)],$$

which is the asserted converse for every $t_0 > 0$. This completes the proof. □

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